

PHYSICS GRADE 10: TEMPERATURE AND THERMAL EXPANSION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (6 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.4: Temperature and Thermal Expansion
Total Duration	6 lessons × 40 minutes = 240 minutes total
Sub-Strand Content	– Temperature as a measure of the degree of hotness or coldness of a body – Thermometric properties and types of thermometers (liquid-in-glass, clinical, thermocouple) – Temperature scales: Celsius, Kelvin, and conversion between scales – Linear expansion of solids: coefficient of linear expansivity (α) and the formula $\Delta L = \alpha L_0 \Delta T$ – Applications and effects of linear expansion (railway tracks, bridges, overhead cables, bimetallic strips) – Superficial (area) expansion of solids and the relationship to linear expansivity – Volumetric (cubical) expansion of solids, liquids, and gases – Anomalous expansion of water and its significance in aquatic ecosystems – Real and apparent expansion of liquids – Applications of thermal expansion (thermostats, riveting, fitting of metal tyres, expansion joints on Kenyan roads and bridges)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - explain temperature as a measure of the degree of hotness or coldness of a body, - describe the working principles of different types of thermometers and their thermometric properties, - convert temperature values between Celsius and Kelvin scales, - determine the linear, superficial, and volumetric expansion of solids using relevant formulae, - investigate the expansion of liquids and distinguish between real and apparent expansion, - explain the anomalous expansion of water and its importance to aquatic life in Kenyan lakes such as Lake Victoria, - apply knowledge of thermal expansion to explain real-life situations such as expansion joints on Kenyan roads and bridges, bimetallic strips in thermostats, and the bursting of sealed containers, - appreciate the role of thermal expansion in engineering design and environmental systems.
Core Competencies	– Communication and Collaboration: developed as learners work in groups to design and conduct experiments on thermal expansion of solids and liquids, share findings, and present conclusions to the class using locally available materials such as metal rods and water-filled bottles. – Critical Thinking and Problem Solving: developed as learners analyse why a sealed glass bottle filled with water cracks when frozen, apply expansion formulae to calculate changes in dimension, and evaluate how engineers account for thermal expansion in construction of roads and railway lines in Kenya. – Learning to Learn: developed as learners independently research applications of thermal expansion in everyday Kenyan

	contexts (e.g., bimetallic strip thermostats, expansion joints on the Nairobi–Mombasa highway) and reflect on how their mental models change across the lesson sequence. – Digital Literacy: developed as learners use digital simulations and videos to visualise molecular behaviour during heating and cooling where physical experiments have limitations, and to record and analyse experimental data.
Core Values	– Responsibility: the learner handles laboratory equipment carefully and cleans up after experiments, demonstrating personal accountability for the shared learning environment. – Respect: the learner listens to and values the contributions of group members when discussing observations on thermal expansion, appreciating that peers may arrive at explanations through different reasoning paths. – Unity: the learner collaborates with classmates from diverse backgrounds during group investigations, recognising that shared effort produces stronger scientific understanding. – Social Justice: the learner reflects on how communities living near Kenyan lakes (e.g., Lake Turkana, Lake Victoria) are affected by temperature-driven water level and density changes, connecting physics to environmental equity.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: embedded in Lesson 1 when learners observe the cracked frozen bottle and generate testable questions about why water expands on freezing and what variables affect the degree of expansion in different materials. – SEP 2 – Developing and Using Models: embedded throughout the unit as learners build and iteratively revise particle-level diagrams and macroscopic expansion models to explain linear, superficial, volumetric, and anomalous expansion. – SEP 3 – Planning and Carrying Out Investigations: embedded in Lessons 2–4 as learners design controlled experiments to measure linear expansion of metal rods and volumetric expansion of water, identifying variables and selecting appropriate measuring instruments. – SEP 4 – Analysing and Interpreting Data: embedded in Lessons 3–4 as learners tabulate, graph, and interpret expansion data, calculating coefficients of linear expansivity and identifying anomalous expansion of water from temperature–volume graphs. – SEP 6 – Constructing Explanations and Designing Solutions: embedded in Lessons 5–6 as learners construct evidence-based explanations for the anchoring phenomenon and propose engineering design solutions (e.g., gap sizing in railway tracks or road expansion joints) that account for thermal expansion. – SEP 8 – Obtaining, Evaluating, and Communicating Information: embedded in Lessons 5–6 as learners research real-world Kenyan applications of thermal expansion, evaluate source credibility, and communicate findings through annotated diagrams and class presentations.
Pertinent & Contemporary Issues (PCIs)	– Environmental Education: learners explore the anomalous expansion of water and its role in sustaining aquatic life in Kenyan freshwater bodies (Lake Victoria, Lake Naivasha), connecting thermal physics to ecosystem health and environmental conservation. – Safety: learners are guided on the dangers of sealing heated containers (e.g., sealed sufuria lids, bottled water left in a hot matatu) and appreciate the importance of pressure-relief design in everyday Kenyan household and industrial contexts. – Life Skills – Critical Thinking and Problem Solving: learners apply thermal expansion concepts to evaluate the safety and efficiency of infrastructure such as the Standard Gauge Railway (SGR) expansion joints and concrete road joints on the Thika Superhighway. – STEM Integration: learners connect physics principles to engineering and construction careers relevant to Kenya's Vision 2030 infrastructure projects, including road building, bridge construction, and industrial manufacturing.
Career Connections	– Civil and Structural Engineer: this sub-strand directly connects to the design of expansion joints in roads, bridges, and buildings — critical skills for Kenya's ongoing infrastructure development under Vision 2030. – Mechanical Engineer: learners see how coefficients of thermal expansion govern the design of engines, pipelines, and

	machinery components used in industries across Nairobi's Industrial Area and Mombasa's port. – Meteorologist / Climatologist: understanding temperature scales and thermal behaviour of materials underpins weather forecasting and climate modelling, relevant to Kenya Meteorological Department careers. – Medical Professional (Clinical Thermometry): the study of thermometers and temperature scales connects to the use of clinical thermometers and patient temperature monitoring in Kenyan hospitals and clinics. – Materials Scientist / Manufacturer: knowledge of expansion properties of solids and liquids is essential for selecting materials in Kenyan manufacturing industries, including metal fabrication and glass production.
Focus for Lessons	This unit investigates how temperature changes cause matter to expand or contract, anchored in the puzzling real-world observation of a water-filled glass bottle cracking after being placed in a freezer. Learners move from qualitative observations of expansion phenomena to quantitative application of linear, superficial, and volumetric expansion formulae, culminating in evidence-based explanations of both ordinary and anomalous thermal behaviour. Throughout the six lessons, Kenyan infrastructure examples — SGR rail joints, Thika Superhighway expansion gaps, bimetallic thermostats in local appliances — ground abstract physics concepts in familiar, meaningful contexts.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature? KICD KEY INQUIRY QUESTIONS: 1. How does temperature affect the physical properties of materials? 2. How is the expansion of solids, liquids, and gases applied in everyday life?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Cracked Frozen Bottle: A learner fills a clean 500 ml glass bottle completely to the brim with water, seals it tightly with a metal cap, and places it in a freezer overnight. The next morning the bottle is cracked — in some cases shattered — and a bulging column of ice protrudes where the cap once sat. This is puzzling because learners' everyday experience tells them that things shrink when they get cold, yet the water clearly expanded enough to crack solid glass. The phenomenon raises immediate questions: Why did the water expand instead of shrink? How much force must the expansion have generated to crack glass? Does all matter expand when cooled? What is special about water? The visual evidence — the precise crack lines running along the bottle, the protruding ice plug, and the undisturbed frozen shape — makes the expansion undeniable and quantifiable, motivating the entire unit's investigation into temperature, thermal expansion, and the anomalous behaviour of water. The teacher presents a physical demonstration bottle (pre-prepared) alongside a labelled cross-section diagram showing: (1) liquid water molecules before freezing, (2) hexagonal ice crystal lattice after freezing with greater intermolecular spacing, (3) the direction of outward pressure on the glass walls, and (4) the resulting crack pattern.
Storyline Thread	Lesson 1 — Encounter and Predict (The Cracked Bottle): Learners observe the pre-prepared cracked frozen bottle and a labelled diagram of the ice crystal structure. They record predictions about why the bottle cracked, whether all materials expand on cooling, and what factors might control the amount of expansion. The class co-constructs the Driving Question Board (DQB) with initial questions and a baseline particle model diagram. Lesson 2 — Temperature Scales and Thermometers (Measuring the "How Hot"): Learners investigate different types of thermometers (liquid-in-glass, clinical, bimetallic strip) and their thermometric properties. They practice converting between Celsius and Kelvin scales using examples from Kenyan weather data (Nairobi vs. Marsabit temperatures). They revisit the DQB to add: "What exactly is temperature measuring at the particle level?" Lesson 3 — Linear Expansion of Solids (The Stretching Rod): Learners conduct or observe a bar-and-gauge experiment using a

	metal rod heated with a Bunsen burner, measuring length change. They derive and apply the formula $\Delta L = \alpha L_0 \Delta T$, solving problems involving railway track gaps on the SGR and overhead cable sag for Kenya Power lines. They update their particle model to show increased vibration amplitude causing bulk expansion. Lesson 4 — Superficial and Volumetric Expansion (From Length to Volume): Learners extend the linear expansion concept mathematically to area ($\beta \approx 2\alpha$) and volume ($\gamma \approx 3\alpha$) expansion, then investigate volumetric expansion of water experimentally using a flask-and-capillary-tube setup. Plotting a temperature–volume graph for water reveals the anomalous expansion below 4 °C — the key to explaining the anchoring phenomenon. Learners directly connect the water-expansion data to the cracked bottle. Lesson 5 — Applications of Thermal Expansion (Engineering Solutions): Learners research and present real Kenyan applications: expansion joints on the Thika Superhighway, bimetallic strip thermostats in electric irons, fitting of metal tyres on wooden cart wheels, riveting in metal roofing. They evaluate how engineers calculate required gap sizes using the linear expansion formula and propose a design solution for a 10 m steel bridge beam in Kisumu exposed to a 35 °C seasonal temperature range. Lesson 6 — Model Revision, Synthesis, and DQB Resolution (Explaining the Crack): Learners return to the anchoring phenomenon with full explanatory power — anomalous expansion of water, volumetric expansion formula, and the pressure generated against glass walls. They revise their cumulative particle model to integrate all lessons, resolve outstanding DQB questions, and produce a final annotated explanation poster connecting the cracked bottle to temperature scales, expansion coefficients, anomalous water behaviour, and real-world engineering design.
Supporting Phenomena	– The sagging overhead power line: Learners observe photographs of Kenya Power transmission lines — visibly slack and sagging on a hot Nairobi afternoon but taut and straight on a cold Kericho highland morning — prompting questions about how temperature affects the length of metal cables. – The stuck metal lid on a glass jar: A learner cannot open a metal lid on a glass jar of jam; running hot water over the lid for 10 seconds makes it open easily, illustrating differential thermal expansion between metal and glass and linking directly to the concept of coefficients of expansivity. – The bimetallic strip in a toaster or iron box: A bimetallic strip (brass and invar) straightens and bends visibly as it is heated and cooled, demonstrating that different metals expand by different amounts and that this difference can be harnessed to do mechanical work in thermostats and automatic switches common in Kenyan households. – Gap in railway tracks: Learners examine photographs of the Nairobi Commuter Rail and SGR tracks, observing deliberate small gaps left between rail sections — gaps that are tighter in the midday heat and wider on cold mornings — providing a compelling engineering application of linear expansion that learners can measure and calculate.

LESSON 1 (80 minutes): Encounter and Predict: The Cracked Frozen Bottle

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon, elicit prior knowledge, and establish the Driving Question Board that will guide the entire sub-strand investigation.
Knowledge	Learners will know that matter exists in three states (solid, liquid, gas) and that temperature is a measure of the average kinetic energy of particles. They will know that water behaves anomalously when it freezes, expanding rather than contracting, and that this expansion generates measurable outward pressure.
Skills	Learners will be able to: (a) observe and describe physical evidence from the cracked frozen bottle phenomenon; (b) construct a baseline particle model diagram representing water molecules before and after freezing; (c) formulate investigable questions and record them on the Driving Question Board; (d) make and justify written predictions about whether all materials expand or contract when cooled.
Attitudes	Learners will appreciate that puzzling everyday observations — such as a cracked bottle in a freezer — are the starting point for rigorous scientific inquiry. They will demonstrate curiosity, open-mindedness, and willingness to revise initial predictions in light of evidence.
Key Inquiry Question	Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?
Purpose in Storyline	This is the opening lesson of the sub-strand. Its sole purpose is to create a productive intellectual disturbance: learners encounter undeniable physical evidence (the cracked bottle, the protruding ice plug, the crack-line pattern) that contradicts their intuitive expectation that things shrink when cooled. By surfacing and recording their initial mental models before any instruction occurs, the teacher establishes a baseline that every subsequent lesson will refine. The Driving Question Board co-constructed today becomes the living record of the class's growing understanding throughout all lessons in Sub-Strand 1.3.
Safety Notes	Handle the pre-prepared cracked glass bottle using thick laboratory gloves and place it inside a clear plastic tray to contain any glass fragments. Do not allow learners to handle the cracked bottle directly. If the bottle is shattered, display it sealed inside a transparent zip-lock bag taped to the tray. Remind learners that broken glass is a puncture hazard and must never be touched with bare hands. If a live freezing demonstration is staged for a later lesson, learners must stand at least one metre from the freezer when it is opened.

B. LESSON OVERVIEW

Learners encounter the anchoring phenomenon — a sealed glass bottle filled with water that cracked overnight in a freezer — through direct observation of the pre-prepared physical specimen and a labelled cross-section diagram. Working individually, then in pairs, they write predictions about why the bottle cracked and whether all materials behave the same way when cooled. The class shares predictions using a structured cold-call protocol and the teacher records them on a T-chart (Shrinks / Expands / Unsure). Learners then sketch a baseline particle model of liquid water and frozen water. The lesson closes with the whole class co-constructing the Driving Question Board, populating it with their most burning questions about temperature, expansion, and the strange behaviour of water. No answers are given in this lesson — the goal is productive uncertainty and a rich DQB that motivates the lessons ahead.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Physical and chemical changes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners enter to find the pre-prepared cracked frozen bottle (inside a clear plastic tray) displayed at the front of the room alongside a labelled A3 cross-section diagram showing: (1) liquid water molecules before freezing, (2) hexagonal ice crystal lattice after freezing with greater intermolecular spacing, (3) the direction of outward pressure on the glass walls, and (4) the resulting crack pattern. Working in silence for 5 minutes, each learner individually completes a printed 'Predict' card with three sentence stems: (a) 'I think the bottle cracked because...', (b) 'When things get cold, I think they generally... because...', (c) 'I predict that if we tested a plastic bottle instead of glass, the result would be... because...'. Learners then share with a partner (Think-Pair) for 3 minutes before the teacher opens a whole-class cold-call share-out.	BEFORE CLASS: Prepare the cracked bottle specimen at least one day in advance by filling a 500 ml glass bottle to the brim, sealing it with a metal cap, and freezing it overnight. Place the specimen in a clear plastic tray at the front bench. Print one A3 labelled diagram per pair and one 'Predict' card per learner. OPENING (0–2 min): Stand beside the specimen without speaking. Allow learners to notice it. After 30 seconds say: 'Take a close look at what is on the bench. Do not touch it yet. In your exercise book, write down three words that describe what you see.' WAIT 60 seconds. PREDICT CARD LAUNCH (2–7 min): Distribute Predict cards. Say: 'Read the three sentence stems. Answer each one in your own words — there are no wrong answers right now. This is your thinking before we investigate. You have 5 minutes. Work in silence.' Set a visible timer. THINK-PAIR (7–10 min): 'Turn to your partner. Share your three predictions. Listen carefully — you will need to report your partner's idea, not your own.' WAIT 3 minutes. COLD-CALL SHARE-OUT (10–18 min): Cold-call exactly 6 pairs (12 learners). For each pair say: 'Tell me your partner's prediction for stem (a).' Record every unique idea verbatim on the board under a T-chart headed: SHRINKS	STRATEGY: Think-Pair-Cold-Call with a T-Chart Prediction Wall. This strategy externalises learners' prior mental models before any instruction occurs, creating a public record of naive conceptions (e.g., 'everything shrinks when cold') that later lessons will directly confront and revise. Recording predictions on a T-chart makes the range of views visible to the whole class, normalises disagreement, and gives the teacher immediate diagnostic information about which misconceptions are most prevalent — especially the dominant misconception that cooling always causes contraction.	OBSERVABLE INDICATOR: Every learner has three written sentences on their Predict card by the end of the 5-minute individual work period. Teacher circulates and visually scans cards — look specifically for learners who leave stem (b) blank or write 'I don't know', as these learners may lack the particle model prerequisite knowledge and will need additional scaffolding in Phase 3. DIAGNOSTIC QUESTION to ask quietly to those learners: 'What do you think happens to the spaces between water molecules when the water gets very cold?'

		WHEN COLD EXPANDS WHEN COLD UNSURE / DEPENDS. After each response ask: 'Does anyone have a different prediction? What is your reasoning?' WAIT TIME: 10 seconds after each question before accepting responses. Avoid affirming or correcting — say: 'Interesting. Let's hold that idea.' If a learner says 'water is weird' or 'water is different', mark it with a star and say: 'That word — different — is going to matter a lot in this unit. Keep it in mind.'		
Observe Phase  VIDEO: Physical and chemical changes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Pairs receive the labelled A3 cross-section diagram and a structured 'Observation Record' sheet with four sections: (1) Describe the crack pattern — where did the glass break first?, (2) Describe the ice plug — what does its shape tell you about the direction of expansion?, (3) Study the molecular diagrams — what differences do you notice between the liquid water arrangement and the ice crystal arrangement?, (4) Record one thing you see that surprises you. Pairs spend 10 minutes completing the Observation Record in silence, then the teacher guides a whole-class gallery-walk style debrief. The teacher also reads aloud a short contextual passage (120 words) describing how Kenyan households in Nairobi and Eldoret have found cracked clay water pots and metal jerricans during unusually cold highland nights — connecting the phenomenon to learners' lived experience.	DISTRIBUTE DIAGRAM AND RECORD SHEET (18–20 min): 'You now have the labelled diagram and the Observation Record sheet. Your job is to be a detective — use only what you can see in the diagram and on the specimen tray to fill in the four sections. Do not use your phone or textbook yet. Evidence only.' WAIT 10 minutes with timer visible. CIRCULATION PROTOCOL (20–28 min): Move to every pair at least once. Ask targeted questions — do NOT give answers: 'What direction are the arrows pointing in panel 3?' 'How does the spacing between molecules in the ice diagram compare to the liquid water diagram?' 'If the molecules moved further apart, what would happen to the total volume?' WAIT 10 seconds after each question. KENYAN CONTEXT BRIDGE (28–30 min): Read aloud: 'In Eldoret and Kericho, where highland temperatures can drop below 4°C in July, some families have found their sealed clay water pots	STRATEGY: Structured Observation with a Labelled Diagram + Kenyan Context Bridge. By providing a pre-labelled cross-section diagram rather than a blank one, the teacher ensures all learners — regardless of prior art or drawing ability — can access the molecular-level evidence. The four-section Observation Record scaffolds noticing from macroscopic (crack lines, ice plug shape) to microscopic (molecular spacing), training the NGSS practice of Analysing and Interpreting Data. The Kenyan context bridge (Eldoret clay pots, Aberdare radiator pipes) activates cultural prior knowledge and signals that the phenomenon is not a laboratory curiosity but an everyday reality in highland Kenya.	OBSERVABLE INDICATOR: In section (3) of the Observation Record, every learner should have written that ice molecules are spaced further apart / have larger gaps than liquid water molecules. If fewer than 70% of pairs have this correct observation during circulation, the teacher pauses the class and directs all learners to look at panel 1 vs panel 2 of the diagram again, asking: 'Count the number of molecules in the same-sized box in panel 1 and panel 2. Which panel has fewer molecules in the same space?' This re-focuses attention on the key evidence without telling learners the explanation.

		cracked by morning. Drivers in the Aberdare highlands have found water-filled radiator pipes split overnight. This is the same phenomenon you are looking at right now.' Ask the class: 'Has anyone here, or in their family, seen something crack or burst because of cold? Put your hand up.' Cold-call 3 learners who raise their hands. Record their examples on the board. WAIT 10 seconds between each example. COLD-CALL DEBRIEF — 4 questions, 4 different learners each time (30–35 min): Q1: 'Learner A, describe the crack pattern — where did the glass break first and why might that be?' Q2: 'Learner B, what does the shape of the ice plug tell us about which direction the water pushed?' Q3: 'Learner C, how many sides does the ice crystal structure in panel 2 have, and what do you notice about the spaces inside it compared to panel 1?' Q4: 'Learner D, what is the one thing on your sheet that surprised you most?' After Q3, say: 'Hold that observation — the spaces inside the ice crystal are going to be the key to everything we learn this unit.'		
Explain Phase  VIDEO: Physical and chemical changes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	The teacher introduces the particle model of matter using a brief guided note-taking frame (half-page template with three labelled boxes: SOLID / LIQUID / GAS). Learners complete the frame as the teacher narrates, drawing particle arrangements and labelling intermolecular spacing and particle motion for each state. Learners then apply the model	GUIDED NOTE-TAKING FRAME (35–45 min): Distribute the half-page template. Say: 'We are going to build a model — a scientific explanation tool. Fill in each box as I narrate. Draw particles as small circles. Use arrows to show how fast they are moving.' Narrate slowly, pausing after each state: 'In a solid, particles are tightly packed in fixed positions — they vibrate	STRATEGY: Verbal Guided Modelling + Immediate Application Sketch. By narrating the particle model without drawing it on the board, the teacher requires every learner to actively construct the model rather than copy it, engaging the NGSS practice of Developing and Using Models. The immediate application sketch forces learners to extend the	OBSERVABLE INDICATOR: During the application sketch, the teacher looks for two categories: (A) learners who draw particles converging monotonically through freezing — this is the most common response and represents the key misconception to be addressed in Lesson 2; (B) learners who draw particles diverging below 4°C — this

[Thermal Energy \(Flexbook\)](#)

Source: CK-12
[Search ARES for similar readings](#)

immediately: they sketch their own version of what happens to water particles as temperature drops from 25°C to 0°C and below, annotating the sketch with arrows showing direction of expansion. Pairs compare sketches and identify one similarity and one difference between their diagrams. The teacher closes the Explain phase by introducing the vocabulary term 'anomalous expansion of water' and asking learners to connect it back to the cracked bottle.

but do not move past each other. Draw 9 circles in a neat grid, with small vibration arrows...' Continue for liquid and gas. WAIT 20 seconds after each state for learners to draw. Do NOT draw on the board — let learners construct their own diagrams from verbal description (NGSS: Developing and Using Models).

APPLICATION SKETCH (45–52 min): 'Now close your notes. On the back of the template, sketch what you think happens to water particles as the temperature drops from 25°C to 0°C and then below 0°C. Show the direction the particles move. Annotate with at least three labels.' WAIT 7 minutes. Circulate — look for learners who draw particles getting closer together all the way to 0°C and below (expected — and correct for most substances). Do not correct yet.

PAIR COMPARISON (52–55 min): 'Compare your sketch with your partner. Find one thing that is the same and one thing that is different. Write both on a sticky note.' Collect sticky notes — these feed directly into the DQB.

CONNECT TO PHENOMENON (55–58 min): Cold-call 1 learner: 'Using the vocabulary term anomalous expansion of water, explain in one sentence why the bottle cracked.' WAIT 15 seconds. After response say: 'We have named the behaviour. Now we need to explain it fully — and that is what the next four lessons will do.' Write 'ANOMALOUS

model to the phenomenon before they have been given the 'correct' answer — this reveals the precise point at which their mental model breaks down (most will draw water particles converging all the way through freezing, which is what most substances do, but is wrong for water). This productive failure creates the conceptual need for the concept of anomalous expansion, which the teacher then names — but does not yet fully explain — at the end of the phase.

indicates stronger prior knowledge. The teacher uses the sticky-note pair comparison to tally the proportion of Category A vs B sketches. If more than 80% are Category A, the teacher notes this for Lesson 2 planning and says to the class: 'Most of you drew the particles getting closer all the way down — that is exactly what happens in almost every substance. Water is the exception. We will find out why next lesson.'

		EXPANSION OF WATER' on the DQB anchor strip.		
Driving Question Board (DQB) Creation  VIDEO: Physical and chemical changes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner writes two questions on separate sticky notes — one question about the cracked bottle phenomenon, one question about any aspect of temperature and matter they are now curious about. Sticky notes are posted on a large class DQB chart (poster paper or whiteboard section) under three column headings: WHAT? (observation questions) WHY? (mechanism questions) SO WHAT? (application questions). The teacher facilitates a 5-minute whole-class clustering exercise where learners group similar questions together. The teacher reads out the driving question of the unit and the class votes on which of their sticky-note questions are most closely connected to it. These are circled in red and labelled 'Priority Questions'. The teacher photographs the DQB for the ARES platform record.	STICKY-NOTE LAUNCH (58–63 min): Distribute two sticky notes per learner (different colours: yellow = WHAT/WHY, pink = SO WHAT application). Say: 'Write one question per note. Question 1: something you are genuinely confused or curious about after seeing the cracked bottle. Question 2: a question about where temperature and expansion matter in real Kenyan life — think about cooking ugali, storing petrol, building roads, metal bridges at Likoni Ferry.' WAIT 5 minutes. Circulate and prompt quiet learners: 'What surprised you most? Turn that surprise into a question.' POSTING AND CLUSTERING (63–68 min): Invite learners row by row to post their notes on the DQB. While they post, say: 'If your question is similar to one already up there, post it right next to it — we are building clusters.' After all notes are posted, point to the largest cluster: 'This cluster has the most questions — what big idea do they all share?' Cold-call 2 learners. WAIT 10 seconds each. DRIVING QUESTION ANCHOR (68–72 min): Read the driving question aloud twice: 'Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?' Ask: 'Which cluster of your questions is most directly answered by this driving question? Point to it.' Walk to that cluster and	STRATEGY: Collaborative DQB Construction with Priority Voting. The DQB is not a display — it is a cognitive tracking tool. The three-column structure (WHAT / WHY / SO WHAT) scaffolds learners into distinguishing between observational, mechanistic, and application-level questions, which maps directly onto NGSS practices of Asking Questions and Constructing Explanations. The priority-voting step gives learners agency and ensures the driving question feels co-owned rather than teacher-imposed. Returning to the DQB at the close of every lesson and ticking answered questions provides visible evidence of growing understanding, which research shows increases learner motivation and metacognitive awareness.	OBSERVABLE INDICATOR: Every learner has posted two sticky notes. The teacher scans for question quality — specifically, look for learners whose questions are purely factual recall ('What is temperature?') and who have not yet connected to the phenomenon. These learners may lack enough observational grounding. During the clustering activity, the teacher notes whether at least one question in the Priority cluster is about the molecular / particle-level mechanism (e.g., 'Why do water molecules take up more space as ice?'). If no such question appears, the teacher adds one as a 'teacher question' card in a contrasting colour, saying: 'I am adding a question that I think will be important for us — you can decide later whether you agree.'

		circle it in red marker. Say: 'These are our Priority Questions. Every lesson in this unit will answer at least one of them. We will return to this board at the end of every lesson and tick off what we now understand.' PHOTOGRAPH (72–73 min): Photograph the DQB and announce: 'This photo goes on our ARES platform page. You can add questions digitally between lessons if new ones come to you tonight.'		
Model Building Phase  VIDEO: Physical and chemical changes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	As the lesson's closing activity, each learner constructs a Baseline Particle Model — a two-panel annotated sketch in their exercise book labelled 'MY MODEL — LESSON 1 (BEFORE INSTRUCTION)'. Panel A shows water molecules in the liquid state inside the sealed bottle. Panel B shows what the learner currently thinks happens to those molecules when the bottle is placed in the freezer overnight. Learners draw arrows to show direction of molecular movement and label: temperature change, volume change, and pressure on the glass walls. They sign and date the model. The teacher explicitly tells learners: 'This is your starting model. It does not have to be correct — it has to be honest. We will update it at the end of every lesson and compare the final model to this one at the end of the unit.'	MODEL BRIEF (73–76 min): Say: 'Open your exercise book to a fresh double page. Write at the top: MY BASELINE PARTICLE MODEL — LESSON 1. Draw a vertical line down the middle. Label the left panel: LIQUID WATER IN BOTTLE (25°C). Label the right panel: AFTER ONE NIGHT IN THE FREEZER.' Display the sentence frame on the board: 'In Panel A I have drawn... because I think... In Panel B I have drawn... to show... The arrow on the glass wall shows... because...' INDIVIDUAL CONSTRUCTION (76–80 min): WAIT 4 minutes for silent individual work. Circulate — do not correct particle arrangements. If a learner draws only the macro level (just the bottle shape with no molecules), prompt: 'Can you zoom in? Draw what you think is happening at the particle level inside the water.' If a learner is stuck, ask: 'What did you notice in the diagram panel 1 vs panel 2? Start with that.'	STRATEGY: Baseline Model Construction with a Temporal Commitment Frame. By explicitly labelling the model 'BEFORE INSTRUCTION' and asking learners to sign and date it, the teacher creates a scientific notebook culture where models are understood to be provisional and revisable — a core epistemic habit of NGSS practice Developing and Using Models. The temporal commitment frame ('we will return to this exact page at the end of the unit') transforms the baseline model from a throwaway exercise into a longitudinal self-assessment tool. Research in science education shows that learners who can directly compare their initial and final models demonstrate deeper conceptual change than those who only produce a final model.	OBSERVABLE INDICATOR: In Panel B of the baseline model, the teacher looks for three things during circulation: (1) Are molecules drawn further apart than in Panel A (desired — shows uptake of the diagram observation)? (2) Are outward-pointing pressure arrows present on the glass walls? (3) Is a temperature label (e.g., 'below 0°C') included? These three elements are the minimum evidence that the learner has begun to connect the macroscopic phenomenon to a particle-level explanation. Learners missing all three elements are flagged for a 5-minute check-in at the start of Lesson 2. The teacher reviews all collected Predict cards and Observation Records that evening and notes the three most common misconceptions to address at the opening of Lesson 2.

		CLOSING COMMIT (80 min): '30 seconds — sign your name and today's date under your model. This is your scientific record. In the last lesson of this unit, we will return to this exact page and you will see how much your model has grown.' Collect Predict cards and Observation Record sheets for teacher analysis overnight. Do NOT collect the exercise books — the model stays with the learner as a running record.		
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D. TEACHER REFLECTION

After this lesson, review the collected Predict cards and Observation Record sheets and ask yourself: (1) What percentage of learners drew molecules moving further apart in their baseline model Panel B? If fewer than 40%, the molecular diagram in Lesson 2 needs more scaffolding time. (2) Were any of the DQB Priority Questions about the application of thermal expansion in Kenyan infrastructure (roads, bridges, railway lines, the SGR)? If not, plan to introduce this context explicitly at the start of Lesson 2 so learners can see the relevance beyond the bottle. (3) Did any learners challenge the 'anomalous' framing and suggest that maybe all liquids expand when frozen? If so, elevate that question — it is scientifically rich and can anchor a peer debate in Lesson 3. (4) How was the pace? The Predict and DQB phases tend to run long if cold-call discussions are not timed. Consider using a visible 10-second sand timer for wait time in future lessons. (5) Note which learners were silent throughout — did they post sticky notes? Did they complete both model panels? These learners need a low-stakes verbal check-in before Lesson 2 begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A pre-prepared sealed glass bottle filled with water that cracked overnight in a freezer, displaying a protruding ice plug and a distinctive crack-line pattern along the glass walls. Learners also examined a labelled A3 cross-section diagram showing liquid water molecules, the hexagonal ice crystal lattice, the direction of outward pressure, and the resulting crack pattern.
What did I learn?	Water behaves anomalously when it freezes — unlike most substances, it expands rather than contracts as it solidifies. This expansion generates sufficient outward pressure to crack a sealed glass bottle. At the particle level, ice molecules in a hexagonal lattice are spaced further apart than liquid water molecules, meaning ice occupies a greater volume than the same mass of liquid water.
How does this explain the phenomenon?	The cracked bottle is explained by the anomalous expansion of water: as water freezes at 0°C, its molecules arrange into a hexagonal ice crystal lattice with larger intermolecular spacing than liquid water. Because the bottle was completely sealed with no room for expansion, the increasing volume of the freezing water generated enormous outward pressure on the glass walls, eventually exceeding the tensile strength of the glass and causing it to crack. This behaviour is anomalous because almost all other substances contract when they cool and solidify.

LESSON 2 (80 minutes): Temperature Scales and Thermometers: Measuring the "How Hot"

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate different types of thermometers and their thermometric properties, and to practice converting between Celsius and Kelvin temperature scales.
Knowledge	By the end of the lesson, the learner should be able to: (a) describe the thermometric properties of liquid-in-glass, clinical, and bimetallic strip thermometers; (b) explain what temperature measures at the particle level; (c) convert temperatures between the Celsius and Kelvin scales using the relationship $T(K) = T(^{\circ}C) + 273$.
Skills	By the end of the lesson, the learner should be able to: (a) read and record temperatures accurately from different thermometer types; (b) perform Celsius-to-Kelvin and Kelvin-to-Celsius conversions using real Kenyan weather data; (c) construct and annotate a comparative diagram of thermometer types.
Attitudes	The learner appreciates the importance of precise measurement tools in scientific investigation and develops curiosity about the particle-level meaning of temperature.
Key Inquiry Question	How does temperature affect the physical properties of materials? How is the expansion of solids, liquids, and gases applied in everyday life?
Purpose in Storyline	In Lesson 1, learners observed the cracked frozen bottle and generated questions about why water expanded when cooled. This lesson equips learners with the precise measurement tools and language — thermometers and temperature scales — needed to quantify the 'how hot' or 'how cold' at the heart of the phenomenon. Learners will understand that the freezer temperature ($\sim -18^{\circ}C / 255 K$) drove the phase change that cracked the bottle, and begin connecting temperature readings to what is actually happening among water molecules. This evidence feeds directly into the Driving Question Board as learners can now ask: 'At what exact temperature does water expand anomalously, and how do we measure that?'
Safety Notes	Handle glass thermometers with care — if a thermometer breaks, do NOT touch the liquid; inform the teacher immediately. Mercury thermometers should NOT be used; use alcohol (red-liquid) thermometers only. Do not place thermometers in boiling water without teacher supervision. Bimetallic strip demonstrations involving heat sources must be conducted by the teacher only. Learners must wash hands after handling thermometers.

B. LESSON OVERVIEW

Learners begin by predicting which of three thermometer types would best measure the freezer temperature that cracked the bottle. They then conduct a structured observation activity — reading temperatures from liquid-in-glass, clinical, and bimetallic strip thermometers in stations, recording data in a comparison table. A guided sensemaking discussion unpacks the thermometric property (a measurable physical property that changes predictably with temperature) behind each instrument. Learners then apply the Celsius–Kelvin conversion to real Kenyan weather data (Nairobi: $18^{\circ}C$; Marsabit: $38^{\circ}C$; freezer: $-18^{\circ}C$), converting each to Kelvin to build intuition about absolute zero. The lesson closes with a DQB update where learners add new questions about particle-level temperature meaning, and a cumulative model revision connecting thermometer readings to the bottle-cracking phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Pressure versus Temperature (PLIX) Source: CK-12 Search ARES for similar readings	Learners are shown three thermometers on the demonstration bench — a red-liquid (alcohol) liquid-in-glass thermometer, a clinical thermometer, and a bimetallic strip thermometer. They individually write a prediction (2–3 sentences in their science notebooks) answering: 'Which of these three thermometers would give you the most reliable reading of the freezer temperature that cracked the bottle, and why?' Learners then share predictions with a partner using a Think-Pair-Share protocol before selected pairs report to the class. Learners also predict: 'What is the lowest temperature any of these thermometers can measure?'	Teacher holds up each thermometer in turn and names it clearly: 'This is a liquid-in-glass thermometer — the red alcohol column rises and falls. This is a clinical thermometer — you have seen nurses use it at Kenyatta National Hospital. This is a bimetallic strip — two metals bonded together that bend when heated.' Teacher then says: 'Before we test anything, I want you to commit to a prediction in writing. You have 3 minutes — go.' [WAIT 3 minutes, circulate, note 2–3 interesting predictions verbatim.] After Think-Pair-Share (2 minutes), cold-call FOUR pairs — two whose predictions favour the liquid-in-glass and two who chose differently — asking: 'Tell us your prediction AND the evidence or reasoning behind it.' Teacher records key prediction claims on the board under the heading 'OUR PREDICTIONS' without affirming or correcting. Teacher connects to phenomenon: 'Remember — the freezer was running at about minus 18 degrees Celsius all night. Our thermometer needs to survive and read accurately at that temperature. Keep that in mind.'	Think-Pair-Share with Prediction Commitment: By requiring written individual predictions before discussion, every learner activates prior knowledge and takes an intellectual stake in the outcome. Recording class predictions publicly creates cognitive dissonance when observations differ, motivating deeper engagement in the Observe phase.	Teacher scans written predictions during circulation. Observable indicators of readiness: (1) learner identifies at least one physical property (e.g., 'the liquid expands') as the basis for prediction; (2) learner references the freezer temperature or the phenomenon unprompted; (3) learner distinguishes between the thermometers by use-case. Learners who write only 'I think the clinical one' with no reasoning are flagged for targeted questioning during station work.
Observe Phase  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML:	Learners rotate through three stations in groups of four, spending approximately 7 minutes at each station. Station A (Liquid-in-Glass Thermometer): Groups measure and record the temperature of (i) a cup of room-temperature water, (ii) a cup of cold water with ice, and (iii) their own palm. They sketch the thermometer and label: bulb, capillary tube, scale, and alcohol	Teacher sets up stations before class and posts a laminated instruction card at each. At the start: 'You have exactly 7 minutes at each station — I will call time. Your job is to observe, record, and ask questions — not to reach conclusions yet. Fill in every cell of your observation table before you rotate.' During Station C, teacher narrates the demo: 'Watch the strip carefully — I am heating the	Structured Observation with a Comparison Table: The three-column table (Thermometer Type Thermometric Property Range/Limitation) is a disciplinary literacy tool that mirrors how scientists categorise instruments. Requiring learners to name the thermometric property — not just describe what they see — bridges observation and scientific vocabulary. The teacher's	Teacher checks observation tables during rotation transitions. Observable indicators: (1) learner correctly identifies a distinct thermometric property for each thermometer (e.g., 'liquid expands,' 'strip bends,' 'constriction holds reading'); (2) learner records at least one limitation (e.g., clinical thermometer cannot read below 35 °C; bimetallic strip gives relative

Pressure versus Temperature (PLIX) Source: CK-12 Search ARES for similar readings	column. Station B (Clinical Thermometer): Groups examine a clinical thermometer, identify the constriction (kink), record the resting reading, and discuss why the constriction is a design feature. They record: 'What stops the mercury/alcohol column from falling back before a nurse reads it?' Station C (Bimetallic Strip — Teacher-Supervised Demo): Teacher heats one end of a bimetallic strip with a candle or spirit lamp; learners observe and record the direction of bending. They match the bending direction to the metal with higher thermal expansion (copper side bends toward, brass side bends away) and sketch before/after diagrams. All observations are recorded in a structured three-column table: Thermometer Type Thermometric Property Range/Limitation.	copper-brass junction now. [WAIT 10 seconds after heating begins.] What do you notice? Call it out.' [Accept 3 verbal responses.] Teacher then asks: 'Which metal do you think is expanding more — the copper or the brass? Write your answer before I tell you anything.' [WAIT 15 seconds.] After all rotations, teacher cold-calls THREE groups — one per station — to share one surprising observation: 'Group two, what was the one thing at Station B that you did not expect?' Teacher records all observations on the board without editorialising. Teacher connects to phenomenon: 'We now have three different instruments, three different thermometric properties. The question is — which one of these properties let us measure minus 18 degrees Celsius reliably enough to understand what happened to our bottle?'	deliberate withholding of answers during the demo (asking learners to commit before being told) sustains inquiry momentum.	not absolute reading); (3) learner's Station C sketch shows the correct direction of bending with a label. Groups whose tables are incomplete are directed: 'You are missing the limitation column — go back and add it before we move on.'
Explain Phase  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Pressure versus Temperature (PLIX) Source: CK-12 Search ARES for similar readings	Learners first consolidate their station observations in a whole-class gallery debrief. Teacher then introduces the concept of thermometric property formally, and learners annotate their notes. The second part of the Explain phase introduces temperature scales. Learners receive a data card showing real Kenyan weather readings: Nairobi (daily average: 18 °C), Marsabit (peak afternoon: 38 °C), Kericho (early morning: 9 °C), and a standard freezer (–18 °C). Working in pairs, learners convert all four readings to Kelvin using $T(K) = T(^{\circ}C) + 273$, then answer three analysis questions: (Q1) Which location's temperature, in Kelvin, is closest to absolute zero? (Q2) What does it mean that	Teacher opens the Explain phase: 'Let us pull your observations together. I heard three different properties — liquid expansion, metal bending, and a constriction mechanism. Scientists call any physical property that changes predictably and measurably with temperature a THERMOMETRIC PROPERTY. Write that definition now.' [WAIT 20 seconds for writing.] Teacher writes on the board: THERMOMETRIC PROPERTY — a physical property that varies continuously, noticeably, and reproducibly with temperature. Teacher then transitions: 'We now have a measurement problem. Our thermometers give us numbers — but on which scale? And why does	Data Analysis with Contextualised Conversion (Anchored Instruction): Using real Kenyan weather data makes the Celsius–Kelvin conversion meaningful rather than procedural. Pairing the arithmetic with Q3's particle-level question forces learners to connect the abstract Kelvin scale to molecular kinetic energy — the conceptual bridge between temperature measurement and thermal expansion. The teacher's insistence on 'kinetic energy' language in answers scaffolds disciplinary vocabulary without simply lecturing.	Teacher collects one paired worksheet per group at the end of the Explain phase. Observable indicators: (1) all four Celsius-to-Kelvin conversions are correct, including the negative value (–18 °C = 255 K); (2) Q3 answer includes reference to particle motion or kinetic energy, not just 'cold water'; (3) learner can articulate why 0 K is a meaningful lower bound in a way that connects to the phenomenon. Common error to flag: learners who write $273 - 18 = 255$ correctly but state '255 K is very close to absolute zero' — these learners need a sense-of-scale intervention (0 K vs. 255 K is still a large difference on the absolute scale).

	the freezer reads 255 K — is that 'close' to absolute zero on the Kelvin scale? (Q3) If absolute zero (0 K) means particles have minimum kinetic energy, what does the difference between Nairobi (291 K) and the freezer (255 K) tell us about the water molecules in the bottle before and after freezing? Pairs then share Q3 answers with a neighbouring pair before a whole-class discussion anchored to the phenomenon.	it matter for our bottle story?' Teacher distributes the Kenyan weather data cards and says: 'Work with your partner. Convert all four temperatures to Kelvin. You have 5 minutes.' [WAIT 5 minutes, circulate, look for arithmetic errors — especially with negative Celsius values.] Teacher then cold-calls FIVE pairs on Q3, specifically seeking the particle-level language: 'I want to hear the word KINETIC ENERGY in your answer — if it is not there yet, add it.' Teacher narrates the connection: 'At 291 Kelvin in Nairobi, water molecules are moving fast and far apart enough to stay liquid. At 255 Kelvin in the freezer, they slow down so much that they lock into a hexagonal crystal lattice — and that lattice takes up MORE space. That is why the bottle cracked.' Teacher draws a quick particle diagram on the board: liquid water molecules (close, disordered, fast arrows) vs. ice lattice (spaced, ordered, slow arrows), referencing the cross-section diagram from Lesson 1.		
Driving Question Board (DQB) Creation  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Pressure versus Temperature (PLIX)	Each learner writes one new question and one new 'I now understand...' statement on separate sticky notes and adds them to the class DQB. New questions are clustered under the emerging theme: 'Temperature at the particle level.' Sample learner-generated questions anticipated: 'If particles slow down when cold, why do water particles arrange into a structure that takes MORE space?', 'Does absolute zero actually exist — has any scientist ever reached it?', 'Would the bottle have cracked at exactly 0 °C (273 K) or only below that?' Learners	Teacher distributes two sticky notes (different colours: one for questions, one for 'I now understand') to each learner. 'You have 3 minutes — write one genuine question the lesson raised for you and one thing you can now explain that you could not explain at the start of today.' [WAIT 3 minutes.] Teacher invites learners to place notes on the DQB. Teacher reads aloud THREE new questions chosen for their scientific depth — specifically ones that anticipate upcoming lessons (anomalous expansion of water, particle kinetic energy theory):	Driving Question Board — Question Clustering and Dot Voting: The DQB functions as a public, cumulative knowledge-tracking tool. Clustering new questions by theme (particle-level temperature) makes the conceptual progression of the unit visible to learners. Dot voting on Lesson 1's question provides metacognitive data — learners assess their own understanding growth — and signals to the teacher which concepts need revisiting.	Teacher reads all sticky notes after class. Observable indicators of quality: (1) 'I now understand' notes reference specific content (e.g., 'I now understand that the Kelvin scale starts at absolute zero and cannot go negative') rather than vague statements ('I understand temperature'); (2) new questions are scientifically generative — they point forward toward particle theory or anomalous expansion; (3) dot votes show that at least 60% of learners claim partial or full understanding of the Lesson 1 DQB question about particle-level

Source: CK-12 Search ARES for similar readings	also revisit the question added in Lesson 1 ('What exactly is temperature measuring at the particle level?') and vote with a dot sticker on whether they can now partially answer it, fully answer it, or still need more evidence.	'This question — why does the ice lattice take more space — is exactly what Lesson 3 will investigate. Keep it visible.' Teacher draws a red arrow on the DQB connecting today's new questions to the anchoring phenomenon image (the cracked bottle photo). Teacher then facilitates the dot-sticker vote on the Lesson 1 question about particle-level temperature, narrating: 'Six of you think you can now partially answer this. Let us hear one of those partial answers.' [Cold-call ONE learner, affirm particle/kinetic energy language, note gaps for Lesson 3.]		temperature.
Model Building Phase  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Pressure versus Temperature (PLIX) Source: CK-12 Search ARES for similar readings	Learners retrieve their cumulative model from Lesson 1 (a sketch showing the bottle, the ice, and arrows indicating outward pressure). They now annotate and revise the model to include: (1) a temperature label on the freezer environment (-18°C / 255 K); (2) a thermometer icon indicating how that temperature was measured and with which instrument; (3) a particle-level inset box showing liquid water molecules (labelled: 'higher kinetic energy, $\sim 291\text{ K}$, Nairobi temperature') alongside ice crystal molecules (labelled: 'lower kinetic energy, 255 K, freezer temperature'); (4) an arrow labelled '$\Delta T = -36\text{ K}$' showing the temperature drop from Nairobi ambient to freezer. Learners share revised models with a partner and identify one new element their partner added that they missed.	Teacher distributes learners' Lesson 1 model sketches (collected and stored). 'Your model from last lesson showed the physical cracking of the bottle. Today you have new tools — temperature values in Kelvin, particle diagrams, and thermometer knowledge. You have 6 minutes to annotate your model with at least THREE new pieces of evidence from today's lesson.' [WAIT 6 minutes, circulate, prompt with: 'Where is the temperature label? Where are the particles? Which thermometer would you use — show it.'] After peer sharing (2 minutes), teacher cold-calls TWO pairs to describe one revision they made: 'Tell us what you added and WHY it makes your model more complete.' Teacher closes by connecting forward: 'Your model now shows WHAT temperature the water was at and HOW we measured it. Lesson 3 will ask — WHY does water at 255 Kelvin form a lattice that is bigger than liquid water? That is the mystery	Cumulative Model Revision (Progressive Model Building): Returning to the same physical model across lessons makes conceptual growth tangible and visible. Each annotation layer (today: temperature labels, Kelvin values, particle insets) corresponds to a new evidence type gathered in this lesson, reinforcing that scientific models are not fixed but refined as evidence accumulates. The peer comparison step surfaces gaps and alternative representations without requiring teacher intervention.	Teacher collects revised models at the end of class. Observable indicators: (1) model includes both Celsius and Kelvin labels for freezer temperature (-18°C / 255 K); (2) particle-level inset box is present and correctly shows more spacing/order in ice than in liquid water; (3) ΔT annotation is present and correctly calculated; (4) a thermometer type is indicated with a brief justification (e.g., 'liquid-in-glass — can read below 0°C'). Models missing particle insets are flagged for a targeted starter activity in Lesson 3.

		that cracked the bottle.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal: (1) During the station rotations, which thermometer type generated the most learner confusion, and what question or re-explanation would you prepare for next time? (2) When learners converted -18°C to Kelvin, how many made the error of subtracting ($273 - 18$) rather than adding ($-18 + 273$)? What quick-fix activity could address this before Lesson 3? (3) Review the DQB sticky notes — are the new learner questions pointing toward anomalous water expansion (the next lesson's focus) or drifting toward unrelated topics? If drifting, how will you refocus the driving question at the start of Lesson 3? (4) Did the particle diagram you drew connect to the Lesson 1 cross-section diagram clearly enough? Could learners trace the line from 'water molecules slow down $\rightarrow$ form hexagonal lattice $\rightarrow$ lattice is larger $\rightarrow$ bottle cracks'? If not, plan a 5-minute bridging starter for Lesson 3. (5) Were all learner groups represented in cold-call responses, or did the same learners dominate? Review your cold-call log and ensure Lesson 3 prioritises voices not yet heard.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	At Station A, learners observed the red alcohol column in the liquid-in-glass thermometer rising when placed in warm water and falling in ice-cold water. At Station B, learners identified the constriction in the clinical thermometer that prevents the column from falling before a reading is taken. At Station C, learners observed the bimetallic strip bending toward the brass side when heated, indicating that copper expands more than brass under the same temperature change.
What did I learn?	Temperature is a measure of the average kinetic energy of particles in a substance. A thermometric property is any physical property that changes continuously, noticeably, and reproducibly with temperature. Temperature in Kelvin is calculated as $T(\text{K}) = T(^{\circ}\text{C}) + 273$. Absolute zero ($0\text{ K} / -273^{\circ}\text{C}$) is the theoretical point at which particle motion is at its minimum. The freezer temperature ($-18^{\circ}\text{C} / 255\text{ K}$) caused water molecules to slow down and arrange into a hexagonal ice crystal lattice — a lattice that occupies more volume than liquid water at the same mass.
How does this explain the phenomenon?	The water in the sealed bottle was at approximately room temperature ($\sim 18^{\circ}\text{C} / 291\text{ K}$) when placed in the freezer. As the freezer cooled the water to -18°C (255 K), the water molecules lost kinetic energy, slowed down, and reorganised into an ordered hexagonal lattice structure. This lattice has greater intermolecular spacing than liquid water, so the ice occupies a larger volume. Because the bottle was sealed and could not expand, the growing ice exerted outward pressure on the glass walls — eventually exceeding the tensile strength of the glass and causing the bottle to crack.

LESSON 3 (80 minutes): Linear Expansion of Solids — The Stretching Rod

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate how solids expand when heated and apply the linear expansion formula to solve real-world engineering problems.
Knowledge	By the end of the lesson, the learner should be able to: (a) state that solids expand when heated and contract when cooled; (b) define the coefficient of linear expansivity (α) and state its SI units (K^{-1}); (c) recall and apply the formula $\Delta L = \alpha L_0 \Delta T$ to calculate length changes in solids.
Skills	By the end of the lesson, the learner should be able to: (a) conduct or observe a bar-and-gauge experiment to measure the linear expansion of a metal rod; (b) record, analyse and interpret experimental data on length change vs. temperature change; (c) solve quantitative problems involving railway track gaps (SGR) and overhead power-line sag (Kenya Power) using $\Delta L = \alpha L_0 \Delta T$.
Attitudes	By the end of the lesson, the learner should be able to: (a) appreciate that precision in measurement matters for engineering safety; (b) demonstrate curiosity about why engineers deliberately leave gaps in structures; (c) show responsibility when handling heated apparatus and open flames.
Key Inquiry Question	How does temperature affect the physical properties of materials? How is the expansion of solids applied in everyday life?
Purpose in Storyline	In Lesson 1, learners observed that the frozen water bottle cracked — evidence that water expanded on cooling. In Lesson 2, they connected particle vibration to thermal expansion in general. In this lesson (Lesson 3 of 6), learners zoom in on SOLIDS specifically, quantifying exactly HOW MUCH a solid expands per degree of temperature change. By deriving and applying $\Delta L = \alpha L_0 \Delta T$, learners now have a precise tool to answer: 'Could the outward force of expanding ice be calculated? How do engineers account for expansion so that structures don't crack like our bottle?' This advances the driving question from qualitative observation to quantitative explanation.
Safety Notes	 SAFETY: (1) Bunsen burners — learners must tie back long hair, tuck in loose clothing, and never leave a lit burner unattended. (2) The metal rod will become HOT — use insulated gloves or tongs only; never touch the rod bare-handed after heating. (3) Retort stands must be clamped firmly to the bench before heating begins. (4) Inform learners of the location of the fire blanket and first-aid kit. (5) Teacher circulates continuously during the heating phase. (6) Allow the rod to cool fully before dismantling apparatus.

B. LESSON OVERVIEW

Learners begin by predicting whether a metal rod will change length when heated and by how much. They then conduct (or observe via teacher demonstration) a classic bar-and-gauge experiment — heating a steel or copper rod with a Bunsen burner and measuring the length change using a linear-expansion apparatus. From their data, learners derive the relationship $\Delta L = \alpha L_0 \Delta T$ and look up coefficients of linear expansivity for common metals. The Explain phase anchors the formula in two Kenyan engineering contexts: (1) the deliberate expansion gaps left between rails on the Standard Gauge Railway (SGR) running from Mombasa to Nairobi, and (2) the visible sag in Kenya Power overhead cables on hot afternoons in Kisumu. Learners solve structured problems in both contexts. They close by updating their cumulative particle model — annotating increased vibration amplitude as the microscopic cause of macroscopic linear expansion — and revisiting the Driving Question Board to record new evidence bearing on the cracked frozen bottle.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners examine two physical objects placed on their desks: (a) a 30 cm metal ruler and (b) a printed photograph of the SGR rail track taken near Voi Station, clearly showing a small gap between two rail sections. Working individually (3 min), each learner writes a prediction in their science notebook responding to the prompt: 'If this steel ruler were heated from 25 °C to 200 °C, predict: (i) would it get longer, shorter, or stay the same? (ii) estimate how many millimetres it might change — give a number, not just a direction. Justify your prediction using what you know about particles.' After individual writing, pairs share predictions (2 min). The teacher cold-calls 4 pairs (varied confidence levels) to share, recording predicted values on the board without comment. The teacher then displays the photo of the SGR rail gap and asks: 'Engineers left this gap deliberately. What problem were they preventing? Write one sentence.' Cold-calls 3 learners. Predictions remain visible on the board throughout the lesson as a reference point for revision.	1. Place ruler and SGR photo on desks BEFORE learners enter so the prompt is immediate. 2. Display the written prediction prompt on the board/screen: 'Longer, shorter, or same? By how many mm? Why?' 3. SAY: 'Write your own prediction first — do not discuss yet. You have 3 minutes. A number is required, not just a direction.' Set a visible timer. 4. After 3 min, SAY: 'Now compare with your partner. Do you agree on the direction? Do you agree on the amount?' Allow 2 min pair talk. 5. Cold-call 4 pairs: 'Amina and Juma — what did you predict and why?' WAIT 12 seconds between calling name and expecting response. 6. Record all predicted values on the board in a column labelled 'OUR PREDICTIONS — Lesson 3'. Do NOT correct any prediction yet. SAY: 'We will return to these numbers at the end of the lesson.' 7. Show SGR photo. SAY: 'This gap is not an accident. In one sentence — what engineering problem does it solve?' WAIT 10 seconds. Cold-call 3 learners. 8. Bridge to observe: 'Let us gather real evidence so we can judge whose prediction was closest.'	ANCHORED PREDICTION (Making Thinking Visible): By requiring a specific numerical estimate before any instruction, learners activate prior knowledge and expose misconceptions (e.g., 'metals don't expand much' or 'it won't change at all'). Recording predictions publicly creates cognitive accountability — learners are motivated to compare their estimate with real data. The SGR image immediately grounds the abstract question in a Kenyan engineering context, signalling that the formula they are about to learn has direct national relevance.	OBSERVABLE INDICATOR: Every learner has a written prediction with a numerical estimate and a particle-level justification in their notebook before the class discussion begins. Teacher scans notebooks during pair-share (circulate to 6 desks minimum). RED FLAG: If more than 30% of learners write 'it will get shorter' (contraction misconception), flag this explicitly: 'Interesting — some of us predict shortening. Let us design our observation to test that idea directly.' This prepares learners to be genuinely surprised or confirmed by evidence.
Observe Phase  VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Groups of 4 conduct (or closely observe teacher demonstration of) the linear-expansion apparatus experiment. APPARATUS: steel or copper rod (~50 cm, known length L_0 recorded to 0.1 mm using a vernier calliper before heating), linear-expansion apparatus with a dial gauge or vernier scale, Bunsen burner, retort stand, thermometer (or temperature probe), heat-resistant mat,	1. Before lighting burners, conduct a 60-second safety check: SAY: 'Show me — hair tied back? Thumbs up. Gloves on the desk and ready? Thumbs up. Rod is clamped — do not touch it with bare hands during or after heating. Agreed?' Wait for visible agreement from all groups. 2. SAY: 'Your job during heating is to be precise scientists. Read the dial gauge and thermometer	DATA COLLECTION WITH REAL APPARATUS (Empirical Evidence Building): The experiment makes thermal expansion directly measurable rather than theoretical — learners produce their own data set that they will use to derive the formula. The structured data table (with a ΔT column pre-included) scaffolds learners toward recognising the proportional relationship before the formula is	OBSERVABLE INDICATOR: Each group's data table shows at least 4 data rows with consistent, increasing ΔL values as temperature rises (confirming expansion, not contraction). Teacher checks: (a) Is L_0 recorded with correct units and precision? (b) Does the ΔT column show correct subtraction? (c) Can learners describe the ΔL – ΔT trend verbally? RED FLAG: If a group's

 HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	insulated gloves, clamp. PROCEDURE: (1) Learners measure and record the exact initial length L_0 of the rod at room temperature T_1 (~25 °C). (2) They heat the rod steadily with the Bunsen burner for 5 minutes, recording temperature at 1-minute intervals. (3) They read the dial gauge at each interval, recording the length change ΔL. (4) After heating stops, they record the maximum temperature T_2 and maximum ΔL. (5) They allow the rod to cool and observe whether it returns to original length. Learners complete a structured data table in their notebooks: Time (min) Temperature T (°C) Length change ΔL (mm) $\Delta T = T - T_1$ (°C) . If apparatus is limited, the teacher conducts the demonstration at the front while all groups record data simultaneously from the projected dial gauge reading. A short (3-minute) video clip of thermal expansion of railway tracks filmed along the Nairobi–Mombasa SGR corridor is shown as supplementary evidence.	simultaneously at each minute. One person reads, one person records, one person watches the flame, one person is the timekeeper. Rotate roles.' 3. Circulate every 90 seconds. At each group, ask: 'What is your current ΔL? What is your current temperature? Is the trend what you predicted?' WAIT 10 seconds per question. 4. After heating: SAY: 'Before we calculate anything, look at your data table. Describe the relationship between ΔT and ΔL in one sentence — is it linear, curved, random?' Cold-call 3 groups. WAIT 12 seconds. 5. Play SGR video clip (3 min). SAY: 'As you watch, write down one thing the engineers in this video know about metals that you are now also discovering.' 6. Cold-call 2 learners after the video: 'What did the engineers know?' WAIT 10 seconds. 7. SAY: 'Your data and this video are now your evidence. In the next phase, we will build the formula that explains both.'	stated. The SGR video clip bridges the laboratory setting to a national infrastructure context, reinforcing that the physics being measured in the classroom is the same physics that Kenyan engineers must account for across 472 km of railway track.	ΔL values are inconsistent or decreasing, check whether the dial gauge is zeroed correctly or whether the rod has slipped from the clamp — address immediately before the Explain phase.
Explain Phase  VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES	PART A — Deriving the Formula (10 min): Using their own data, learners calculate the ratio $\Delta L / (L_0 \times \Delta T)$ for each data row and observe that the value is approximately constant. The teacher introduces this constant as α (the coefficient of linear expansivity), leading to: $\Delta L = \alpha L_0 \Delta T$. Learners copy the formula, units (α in K^{-1}), and a reference table of α values for steel ($11 \times 10^{-6} K^{-1}$), copper ($17 \times 10^{-6} K^{-1}$), aluminium ($23 \times 10^{-6} K^{-1}$), and glass ($9 \times 10^{-6} K^{-1}$) into their notebooks. They use their own	1. SAY: 'Calculate ΔL divided by (L_0 times ΔT) for each row of your table. What do you notice about the answers?' WAIT 15 seconds. Cold-call 3 groups. Write calculated α values on the board. SAY: 'This nearly-constant value is what we call the coefficient of linear expansivity. Its symbol is α.' 2. Write the formula clearly: $\Delta L = \alpha L_0 \Delta T$. SAY: 'Each symbol tells a story — ΔL is how much it stretched, L_0 is how long it started, ΔT is how much the temperature changed, and α tells you how expansive the material is. Steel is	FORMULA DERIVATION FROM OWN DATA (Inductive Reasoning + Quantitative Modelling): Rather than presenting $\Delta L = \alpha L_0 \Delta T$ as a given formula, learners discover α as a constant ratio emerging from their own experimental data. This inductive pathway builds genuine understanding — the formula is not arbitrary but is a compact description of a pattern learners have already seen. The two-problem sequence (SGR then Kenya Power) applies the formula at escalating complexity: the first is purely numerical, the second	OBSERVABLE INDICATORS: (a) Learners can correctly identify L_0, ΔT, and α from a word problem and substitute without unit errors — teacher checks at least 5 notebooks during problem-solving. (b) SGR answer: $\Delta L = 11 \times 10^{-6} \times 25 \times 45 = 0.012375 \text{ m} \approx 12.4 \text{ mm}$ — teacher marks this as the benchmark; accept 12–13 mm range. (c) Kenya Power answer: $\Delta L = 23 \times 10^{-6} \times 200 \times 55 = 0.253 \text{ m} \approx 25.3 \text{ cm}$. (d) For the explanation question: acceptable answers must include 'extra length has nowhere to go' and 'sag is

for similar readings	experimental data to estimate α for their rod and compare with the accepted value. PART B — SGR Problem (7 min): Learners solve in pairs: 'Each steel rail on the SGR between Mombasa and Nairobi is 25 m long at a laying temperature of 20 °C. The maximum rail temperature on the Tsavo plains reaches 65 °C. Calculate: (i) the expansion ΔL of one rail, (ii) the minimum gap that must be left between rails to prevent buckling. ($\alpha_{\text{steel}} = 11 \times 10^{-6} \text{ K}^{-1}$).' PART C — Kenya Power Problem (8 min): Learners solve individually: 'A Kenya Power overhead aluminium cable strung between two pylons near Kisumu is 200 m long at 15 °C. On a hot afternoon the cable temperature rises to 70 °C. (i) Calculate the increase in cable length. (ii) Explain in one sentence why this increased length causes the cable to sag and why engineers design for this sag rather than stringing cables tightly. ($\alpha_{\text{aluminium}} = 23 \times 10^{-6} \text{ K}^{-1}$).' Learners compare answers with a partner and resolve discrepancies before the teacher cold-calls solutions.	stubborn — small α. Aluminium is more generous — bigger α.' 3. Distribute the α reference table. SAY: 'Find the accepted α for your rod material. How close was your experimental value? Calculate your percentage error.' 4. For the SGR problem: SAY: 'You have the formula. You have the numbers. Work in pairs — you have 7 minutes. Show every substitution step.' Circulate to 5 pairs. Do NOT give answers — ask: 'What is your L_0 here? What is your ΔT? Have you checked your units?' 5. Cold-call 1 pair to present their SGR solution on the board. SAY to class: 'Does everyone agree? If you got a different answer, where did the working diverge?' WAIT 12 seconds. 6. For the Kenya Power problem: SAY: 'Now work individually — this is your chance to show me you can do this without a partner. 8 minutes.' Circulate actively. After 8 min, cold-call 1 learner for part (i) and a DIFFERENT learner for part (ii). SAY: 'The second part asks you to EXPLAIN, not calculate. A tight cable on a hot day — what happens? Why do engineers accept the sag?' WAIT 12 seconds before accepting the response. 7. CONNECT TO PHENOMENON: SAY: 'We have just calculated that a 25 m steel rail expands by about 12 mm on a hot day. Now think about our frozen water bottle. Ice has a much larger volume expansion. What does that tell us about the forces involved?' WAIT 10 seconds. Cold-call 2 learners.	requires conceptual explanation, reinforcing that equations describe physical reality. Both problems are anchored in Kenyan national infrastructure, making the physics personally and economically relevant.	designed to prevent snapping or structural damage.' RED FLAG: If learners use $\Delta T = T_2$ instead of $T_2 - T_1$, pause the class and re-teach the meaning of Δ notation before proceeding.
Driving Question Board (DQB) Creation	Each learner receives two sticky notes (different colours: one for NEW EVIDENCE, one for NEW QUESTIONS). On the NEW	1. Distribute sticky notes. SAY: 'You have 3 minutes — write one piece of evidence from today that helps explain the cracked bottle,	DRIVING QUESTION BOARD UPDATE (Knowledge Tracking + Metacognition): The DQB is not a display item — it is a living	OBSERVABLE INDICATOR: The NEW EVIDENCE sticky notes should reference specific quantitative content (e.g., 'Steel

VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Thermal Energy (Flexbook) Source: CK-12 Search ARES for similar readings	EVIDENCE note, they write one specific quantitative or qualitative finding from today's lesson that relates to the driving question about the cracked frozen bottle. On the NEW QUESTIONS note, they write one new question this lesson has raised. Learners post their notes on the class DQB under the appropriate columns. The teacher reads 4–5 notes aloud, facilitating a 3-minute whole-class discussion to cluster the evidence notes into two emerging themes: (A) 'Solids expand when heated — quantifiably' and (B) 'Expansion forces can be large enough to damage structures.' The teacher explicitly links Theme B back to the phenomenon: 'The bottle cracked because expanding ice generated enormous outward pressure — just as expanding rails would buckle the track if there were no gaps. Same physics, different material.' The teacher previews Lesson 4: 'We have now quantified expansion in solids. Next lesson we investigate liquids — does water expand the same way? Is there anything unusual about water near 0 °C?' This preview is written on the DQB as a bridge question.	and one new question today raised for you. Be specific — no vague answers like 'metals expand'.' Set timer. 2. Direct learners to post notes on the DQB (pre-labelled columns: EVIDENCE QUESTIONS EXPLANATIONS IN PROGRESS). 3. Read 4–5 notes aloud. SAY after each evidence note: 'How does this connect to the bottle cracking?' WAIT 10 seconds. 4. Cluster notes visually. SAY: 'I see two groups of evidence forming — expansion is measurable, AND expansion forces can damage hard structures. Who agrees? Who would group these differently?' Cold-call 2 learners. 5. SAY: 'Write the bridge question in your notebooks: Does water expand linearly like a metal rod — or is it different? We will find out in Lesson 4.' 6. Leave the DQB visible and intact for Lesson 4.	epistemic record of the class's growing understanding. By requiring learners to connect today's evidence explicitly to the driving question, the teacher prevents lessons from feeling like isolated topics. The clustering activity models scientific practice: identifying patterns across multiple data points and recognising where knowledge is still incomplete. The bridge question creates intellectual anticipation for Lesson 4.	expands 12 mm per 25 m rail per 45 °C rise' or 'α for steel is $11 \times 10^{-6} \text{ K}^{-1}$') rather than vague generalisations. Teacher reads all posted notes after class to identify misconceptions or incomplete connections — these inform the opening of Lesson 4. RED FLAG: If the majority of evidence notes say only 'metals expand when hot,' learners have not connected the magnitude of expansion to the structural damage in the phenomenon — address this as an explicit whole-class prompt before closing.
Model Building Phase VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners retrieve their cumulative particle model diagram begun in Lesson 1 (showing water molecules before and after freezing) and extended in Lesson 2 (showing particle vibration in solids). Today they add a NEW PANEL specifically for solid linear expansion: a side-by-side drawing of a metal rod at low temperature (T_1) and the same rod at high temperature (T_2). In the T_1 panel,	1. SAY: 'Take out your cumulative model. Today you are adding Panel 3 — linear expansion of a solid. You have 7 minutes. The model must show particles, vibration, spacing, and the macroscopic result. Label everything.' Set timer. 2. After 5 minutes, SAY: 'Add your one-sentence model statement at the bottom of Panel 3. It must include the word 'amplitude,' the word	CUMULATIVE MODEL REVISION (Progressive Model-Building): The model is not redrawn from scratch each lesson — it grows. This cumulative structure forces learners to maintain coherence across lessons: each new panel must be consistent with previous panels. Adding the formula to the diagram bridges the visual/conceptual model with the mathematical model, reinforcing	OBSERVABLE INDICATOR: Panel 3 of the model must contain: (a) two side-by-side diagrams (T_1 vs. T_2) with visibly different lattice spacing and arrow sizes, (b) the annotation 'increased KE → larger amplitude,' (c) the formula $\Delta L = \alpha L_0 \Delta T$ written as a label, and (d) a one-sentence model statement. Teacher does a rapid gallery walk (remaining 3 minutes of lesson), placing a small ✓ on complete

 HTML: Thermal Energy (Flexbook) Source: CK-12 Search ARES for similar readings	particles are shown in a regular lattice with small double-headed arrows representing low-amplitude vibration. In the T₂ panel, the lattice spacing is slightly greater, the double-headed arrows are longer (larger vibration amplitude), and the rod outline is visibly longer. Learners annotate: (1) 'Increased KE → larger vibration amplitude,' (2) 'Greater average separation between particles,' (3) 'Macroscopic effect: $\Delta L = \alpha L_0 \Delta T$,' and (4) 'This is why the SGR has gaps and Kenya Power cables sag.' Learners then write a one-sentence model statement: 'When a solid is heated, its particles vibrate with greater amplitude, increasing the average distance between them, causing the solid to expand in all dimensions — and in a rod, the measurable length increase is given by $\Delta L = \alpha L_0 \Delta T$.'	'separation,' and the formula.' 3. After 7 minutes, display a teacher model diagram on the board. SAY: 'Compare your model with mine. What did I include that you missed? What did you include that I missed? Add or revise yours now.' Allow 2 minutes for revision. 4. Cold-call 1 learner: 'Read your model statement aloud.' WAIT 10 seconds after the reading. Ask the class: 'Does this statement accurately describe what we observed today? Yes or no — show thumbs.' 5. FINAL CONNECTION: SAY: 'Look at your model. The ice in the frozen bottle expanded because its particles rearranged into a hexagonal lattice with MORE spacing — not just vibrated more. That is DIFFERENT from what we modelled today for a metal rod. That difference is what we will investigate in Lesson 4.' Write on the board: 'ANOMALOUS EXPANSION OF WATER — coming in Lesson 4.'	that both describe the same physical reality. The teacher's deliberate comparison of the metal rod model with the ice-in-bottle phenomenon plants the seed for Lesson 4's anomalous water expansion, sustaining narrative tension in the storyline.	models and a dot on incomplete ones — learners with a dot must complete for homework. RED FLAG: If learners draw particles moving faster (translational motion) rather than vibrating with greater amplitude, they are confusing solid and gas models — correct this individually during the gallery walk and address with the whole class at the start of Lesson 4.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your teaching journal:

1. **FORMULA DERIVATION:** Did learners genuinely discover α as a constant from their own data, or did they feel they were reverse-engineering a formula they already suspected? If the latter, how could the data collection be redesigned to make the constancy of α more surprising and authentic?
2. **PROBLEM-SOLVING QUALITY:** On the SGR and Kenya Power problems, what proportion of learners correctly calculated ΔT as a difference (not a final temperature)? If the ΔT error was common, plan a 5-minute warm-up for Lesson 4 using a simple temperature-change question to consolidate this.
3. **PHENOMENON CONNECTION:** Were learners able to articulate — in their own words — how today's lesson about solid expansion connects to the cracked frozen bottle? Or did the two feel like separate topics? If disconnected, adjust the Phase 3 closing question to be more explicit about the force magnitudes involved in ice expansion vs. metal expansion.
4. **DQB QUALITY:** Were the evidence notes specific and quantitative, or vague and general? If vague, consider modelling a high-quality evidence note at the start of Phase 4 and asking learners to rate it before writing their own.
5. **SAFETY:** Were all learners compliant with the Bunsen burner safety protocol? Record any incidents or near-misses and report to the laboratory technician. If safety compliance was poor, consider moving to a teacher-only demonstration format for the heating phase in future lessons.
6. **PACING:** The lesson is designed for 80 minutes across 5 phases. Which phase ran over time? The most common pressure point is Phase 3 (Explain) — if problems took longer than planned, consider assigning the Kenya Power problem as a structured homework task with worked-solution feedback at the start of Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners heated a metal rod using a Bunsen burner and measured its length change using a linear-expansion apparatus with a dial gauge, recording temperature and length change data at 1-minute intervals. They also viewed a video clip of thermal expansion gaps on the SGR railway between Mombasa and Nairobi.
What did I learn?	Learners derived the coefficient of linear expansivity (α) from their own data as the constant ratio $\Delta L / (L_0 \Delta T)$, and applied the formula $\Delta L = \alpha L_0 \Delta T$ to calculate expansion in SGR steel rails ($\Delta L \approx 12.4$ mm per 25 m rail over a 45 °C rise) and Kenya Power aluminium overhead cables near Kisumu ($\Delta L \approx 25.3$ cm per 200 m cable over a 55 °C rise). They updated their particle model to show that heating increases vibration amplitude and average particle separation.
How does this explain the phenomenon?	When a solid is heated, its particles gain kinetic energy and vibrate with greater amplitude, increasing the average distance between neighbouring particles. This microscopic change produces a measurable macroscopic increase in length described by $\Delta L = \alpha L_0 \Delta T$. The same principle — expansion generating force against a rigid boundary — explains why the sealed glass bottle cracked when the water inside froze and expanded, and why SGR engineers must leave deliberate gaps between rails to prevent track buckling.

LESSON 4 (80 minutes): Superficial and Volumetric Expansion — From Length to Volume

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To extend learners' understanding of thermal expansion from one dimension (linear) to two dimensions (superficial/area) and three dimensions (volumetric), and to experimentally investigate the anomalous volumetric expansion of water below 4 °C using a flask-and-capillary-tube setup.
Knowledge	Learners will know that: (a) superficial (area) expansivity $\beta \approx 2\alpha$, and volumetric (cubic) expansivity $\gamma \approx 3\alpha$, where α is the linear expansivity of a material; (b) the formula for volumetric expansion is $\Delta V = \gamma V_0 \Delta T$; (c) water behaves anomalously — it expands as it cools below 4 °C, reaching maximum density at 4 °C; (d) this anomalous expansion of water is the direct cause of the cracked frozen bottle in the anchoring phenomenon.
Skills	Learners will be able to: (a) derive the relationships $\beta \approx 2\alpha$ and $\gamma \approx 3\alpha$ mathematically from the linear expansion formula; (b) set up and use a flask-and-capillary-tube apparatus to measure volume changes of water with temperature; (c) plot and interpret a temperature–volume graph for water, identifying the anomaly below 4 °C; (d) solve quantitative problems using $\Delta V = \gamma V_0 \Delta T$.
Attitudes	Learners will appreciate that: (a) anomalies in nature are not errors but gateways to deeper scientific understanding; (b) careful experimental observation — as modelled by Kenyan and global scientists — is essential to revealing truths that contradict everyday intuition; (c) the unique properties of water have profound consequences for life, ecosystems like Lake Victoria, and everyday objects like glass bottles.
Key Inquiry Question	How does temperature affect the physical properties of materials? How is the expansion of solids, liquids, and gases applied in everyday life?
Purpose in Storyline	In Lessons 1–3, learners established that solids expand linearly when heated and quantified this using the linear expansivity coefficient α . Lesson 4 scales this understanding to two and three dimensions, then pivots to water — the substance at the heart of the anchoring phenomenon. By plotting real experimental data showing water's volume increasing as temperature drops below 4 °C, learners gain the decisive evidence needed to explain why the sealed bottle cracked: water's anomalous expansion generated enough outward pressure to shatter glass. This lesson is the conceptual turning point of the evidence-gathering sequence.
Safety Notes	SAFETY NOTES: (1) Hot water and Bunsen burners — learners must use heat-resistant gloves when handling heated flasks; no loose clothing near open flame. (2) Glass capillary tubes are fragile — handle with care; report any breakage immediately and do not attempt to pick up glass shards with bare hands. (3) If using ice, surfaces may become slippery — keep work area dry. (4) Thermometers (mercury-free preferred) — if a thermometer breaks, inform the teacher immediately; do not touch any liquid. (5) Learners should wear safety goggles during the heating portion of the experiment.

B. LESSON OVERVIEW

In this 80-minute lesson, learners first use guided mathematical derivation to extend linear expansivity (α) to superficial expansivity ($\beta \approx 2\alpha$) and volumetric expansivity ($\gamma \approx 3\alpha$), solving problems grounded in Kenyan engineering contexts (e.g., the expansion joints on the Nairobi Expressway bridge decks). They then conduct a hands-on experiment using a round-bottomed flask filled with water, fitted with a one-hole stopper and a narrow glass capillary tube, heating and cooling the water while tracking the water column height as a proxy for volume change. Temperature–volume data are recorded and graphed. The resulting curve reveals water's anomalous behaviour: volume decreases as temperature rises from 0 °C to 4 °C, then increases above 4 °C — meaning water is densest at 4 °C and expands when frozen. Learners explicitly connect this graph to the cracked bottle, updating their Driving Question Board with new explanatory evidence and revising their class consensus model to show outward

pressure from ice crystal formation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners are shown two scenarios on the board: (A) a 500 ml metal sufuria filled with water and heated on a jiko from 20 °C to 80 °C, and (B) the same sufuria filled with water and cooled from 20 °C to 0 °C. Each learner independently sketches two bar graphs in their notebooks — one showing predicted volume change for scenario A, one for scenario B — then writes one sentence explaining their prediction for each. Pairs then compare sketches and negotiate a shared prediction to report. The teacher cold-calls 4 pairs (2 per scenario) to share predictions, recording key words on the board under two columns: 'Heating' and 'Cooling'. The teacher also holds up the pre-cracked bottle from the anchoring phenomenon and asks: 'Based on what you predicted for cooling, should this bottle have cracked — or not?' Learners record their current position on a sticky note (AGREE / DISAGREE / UNSURE) and post it on the class tally chart.	Open with: 'We have spent three lessons measuring how solids stretch when heated. Today we go further — we scale that up to area, to volume, and then we test water itself.' Display the two sufuria scenarios as a labelled diagram. Say: 'Independently, no talking yet — sketch your predicted volume-change bar graphs and write one sentence each. You have 3 minutes.' [Set visible 3-minute timer.] After individual work: 'Now turn to your partner — compare your graphs. If you disagree, negotiate one shared prediction. 2 minutes.' [Circulate, listen for misconceptions — particularly learners who predict water shrinks uniformly when cooled.] Cold-call 4 pairs: 'Pair at the back window — tell us your heating prediction and your reasoning.' [WAIT 12 seconds after each question before accepting an answer.] Record key vocabulary on board (contract, expand, volume, density). Hold up the cracked bottle: 'You predicted cooling makes water shrink. But this bottle disagrees with you. Write on your sticky note — AGREE the bottle should have cracked, DISAGREE it should not have cracked, or UNSURE.' Collect sticky notes on tally chart — do not correct yet. Say: 'By the end of today, every one of you will be able to explain exactly why this bottle looks the way it does.'	Predict–Observe–Explain (POE) Launch: By eliciting explicit predictions before any instruction, the teacher surfaces the intuitive misconception that water always contracts when cooled. The sticky-note tally makes this misconception visible and public, creating cognitive dissonance that motivates the subsequent observation phase. Recording predictions also gives learners a concrete reference point to revise against after the experiment.	Observable indicator: Are learners' bar graphs showing a uniform decrease in volume for both heating (incorrect — should show increase) and cooling (partially correct for most substances, but anomalous for water below 4 °C)? The teacher scans notebooks during pair-share to identify learners holding the 'water always shrinks when cooled' misconception. Sticky-note tally is photographed and retained — teacher compares it to an identical tally collected at lesson end to measure conceptual shift. Target: at least 70% of sticky notes should move from DISAGREE/UNSURE to AGREE with correct reasoning by lesson end.

Observe Phase  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	PART A — Mathematical Extension (20 min): Working in groups of 4, learners receive a printed 'Derivation Card' that guides them step-by-step to extend linear expansion to area and volume. Starting from $L = L_0(1 + \alpha\Delta T)$, they expand a square of side L_0 to show the new area $A = L_0^2(1 + \alpha\Delta T)^2 \approx L_0^2(1 + 2\alpha\Delta T)$ for small α, thus $\beta \approx 2\alpha$. They repeat for a cube to show $\gamma \approx 3\alpha$. Groups then solve two contextual problems: (i) A rectangular concrete slab on the Nairobi Expressway is 12 m $\times$ 4 m at 15 °C; if $\alpha = 12 \times 10^{-6} \text{ K}^{-1}$, what is the increase in area when the slab reaches 45 °C during the midday sun? (ii) A 2-litre aluminium sufuria ($\gamma = 3 \times 24 \times 10^{-6} = 72 \times 10^{-6} \text{ K}^{-1}$) is heated from 20 °C to 100 °C; by how much does its internal volume increase? Groups record derivations and answers on A3 whiteboards and hold them up for gallery comparison. PART B — Flask-and-Capillary Experiment (30 min): Each group sets up a 250 ml round-bottomed flask filled to the brim with coloured water (food dye added for visibility), sealed with a one-hole rubber stopper through which a 30 cm glass capillary tube protrudes. The initial water column height in the capillary is measured at room temperature (~20 °C) and recorded. Groups then place the flask in a water bath and systematically cool it using ice, recording the capillary column height at 18 °C, 15 °C, 12 °C, 10 °C, 8 °C, 6 °C, 4 °C, 2 °C, and 0 °C (or as low as possible). They then allow the flask to warm and record measurements as temperature rises back through the	PART A: Distribute Derivation Cards. Say: 'You are not just memorising $\beta = 2\alpha$ — you are going to prove it yourselves using algebra you already know. Work as a group; every member must be able to explain every line.' [Circulate actively; when a group is stuck on the binomial expansion step, prompt: 'What happens to the $\alpha\Delta T$ squared term when α is a very small number like 10^{-5}? Can you ignore it?'] [WAIT 10 seconds after prompting before providing further scaffolding.] After 10 minutes: 'Hold up your whiteboards — let's do a gallery check.' Cold-call 2 groups to narrate their derivation aloud. Affirm correct steps; redirect errors by asking the class: 'Does everyone agree with that step? Thumbs up or sideways.' Transition: 'Now let us test not steel or concrete but the star of our mystery — water.' PART B: Demonstrate the flask setup at the front for 2 minutes before groups begin. Say explicitly: 'The capillary tube is your measuring instrument — the higher the water column, the larger the volume. Watch the column, not the flask.' Assign roles within groups: Data Recorder, Thermometer Reader, Ice Manager, Timer. Every 5 minutes circulate and ask one group member at random: 'What is your column height right now — is it going up or down compared to your last reading?' [WAIT 12 seconds.] At the 4 °C data point (or approaching it), pause the whole class: 'Freeze — everyone look at your column. Is it going up or down as you cool toward 4 degrees? Remember your prediction from Phase 1.' [Allow 15 seconds of silent observation.]	Graduated Complexity + Data-from-Evidence: Part A uses structured derivation (rather than direct instruction) so learners construct the $\beta = 2\alpha$ and $\gamma = 3\alpha$ relationships themselves — deepening retention and allowing immediate application to Kenyan engineering contexts. Part B applies the SEP 'Planning and Carrying Out Investigations' and 'Analyzing and Interpreting Data' — learners generate first-hand evidence of the water anomaly rather than receiving it as a claim. The capillary tube acts as an amplifier of tiny volume changes, making the abstract measurable. Plotting the curve themselves ensures learners encounter the anomaly as a genuine surprise in their own data.	PART A: Teacher scans whiteboards for correct derivation steps — specifically whether learners correctly dropped the $(\alpha\Delta T)^2$ term. Indicator of mastery: group can articulate WHY the squared term is negligible. For the sufuria problem, check whether learners used $\gamma = 72 \times 10^{-6} \text{ K}^{-1}$ correctly. PART B: Each group's data table must show at least 6 temperature–height data pairs. The critical observable: does the capillary column rise (volume increase) as temperature drops below 4 °C? Teacher checks each group's graph before Phase 3 begins — a correctly drawn curve will show a minimum at approximately 4 °C (column lowest = volume smallest = densest). Any group whose graph shows a monotonically decreasing column is prompted: 'Look at your data between 6 °C and 2 °C — which direction is the column moving?'
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	same points. Data are recorded in a shared table: Temperature (°C) Capillary Height (mm) Relative Volume Change (increase/decrease). Learners plot Temperature (x-axis) vs. Capillary Height (y-axis) on graph paper, drawing a best-fit curve. PART C — Video Corroboration (5 min): If equipment limitations prevent reaching 0–4 °C range, the teacher plays a 3-minute curated video clip showing the classic water-anomaly experiment, pausing at the 4 °C minimum-volume point and asking learners to annotate their graphs.	PART C: Before playing video, say: 'As you watch, put a star on your graph wherever the video confirms your experimental data.' After video: 'How many of you saw your graph validated by the video? Hands up.'		
Explain Phase  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use their experimental graphs and the molecular-level cross-section diagram (introduced in the anchoring phenomenon) to construct a written explanation in their notebooks answering three scaffolded questions: (1) 'At what temperature did water have its smallest volume (highest capillary column minimum)? What does this mean for density?' (2) 'Using the formula $\Delta V = \gamma V_0 \Delta T$ and $\gamma_{\text{ice}} \approx 3 \times 51 \times 10^{-6} \text{ K}^{-1} = 153 \times 10^{-6} \text{ K}^{-1}$, calculate the volume increase when 500 ml of water at 4 °C freezes to 0 °C. Compare this to the volume of the original bottle — was there room for expansion?' (3) 'Using the molecular diagram, explain in two sentences why ice occupies more space than liquid water at the same mass.' Groups then participate in a whole-class 'Evidence Line' discussion: the teacher draws a horizontal line on the board labelled 'SHRINKS when cooled' on the left and 'EXPANDS when cooled' on the right. Each group places a sticky arrow on the line to show where water sits	Distribute the three scaffolded questions on a half-sheet. Say: 'Use your graph, your molecular diagram, and your formula sheet — not your neighbour, not yet. Individual writing for 6 minutes.' [Set timer.] [Circulate, reading over shoulders. For Q2: if a learner is confused about which γ to use, prompt: 'Are we expanding liquid water or ice here? Check your notes from Lesson 3.'] After 6 minutes: 'Now share your Q1 answer with your partner — do you agree on the temperature of maximum density?' Cold-call 3 individuals on Q1, then 2 on Q2 (ask them to narrate their calculation step by step). For Q3, say: 'I want molecular language — use the words lattice, spacing, and hexagonal.' [WAIT 15 seconds after posing Q3.] Conduct the Evidence Line: 'Each group, send your representative to place your sticky arrow. You must say one sentence that starts with: Our data shows...' After all groups place arrows: 'Notice — every group's arrow is on the right side of this	Claim–Evidence–Reasoning (CER) Framework + Evidence Line: CER structures learners' explanations so that claims are explicitly linked to their own experimental data (not just memorised facts) and to molecular-level reasoning. The Evidence Line is a physical, visible consensus-building tool that mirrors how scientific communities reach agreement — reinforcing the Nature of Science (NOS) idea that scientific knowledge is built on converging evidence from multiple investigators. The scaffolded questions sequence from data reading → quantitative calculation → molecular explanation, moving learners up Bloom's taxonomy within a single phase.	Observable indicators: (1) In Q2, learners should calculate $\Delta V \approx 153 \times 10^{-6} \times 0.5 \times 4 \approx 0.000306 \text{ L} \approx 0.3 \text{ ml}$ from the formula — and recognise that the actual anomalous expansion of water ($\approx 9\%$ volume increase on freezing) is far larger, because γ_{ice} from the formula underestimates the structural expansion; this discrepancy is itself a teachable moment. (2) CER quality rubric (3 points): Claim correctly identifies expansion below 4 °C (1 pt); Evidence cites a specific temperature–height data point from their graph (1 pt); Reasoning mentions hexagonal lattice and increased intermolecular spacing (1 pt). Teacher collects CER frames at end of lesson as exit evidence. (3) Evidence Line: all groups should place arrows on the 'EXPANDS' side — if any group is on the wrong side, probe their data table before Phase 4.

	below 4 °C, and justifies their placement with one piece of data from their graph. Finally, learners individually complete a 'Claim–Evidence–Reasoning' (CER) frame in their notebooks: Claim — 'Water expands / contracts below 4 °C because...'; Evidence — one data point from their graph; Reasoning — the molecular explanation linking hexagonal ice lattice to greater intermolecular spacing.	line for below 4 °C. That is not a coincidence — that is a scientific consensus built from your own data.' Transition to CER: 'Now write your CER individually. This is your explanation of the cracked bottle — you are the scientist filing the report.'		
Driving Question Board (DQB) Creation  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board: 'Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?' Each group is assigned one of four DQB zones to update: (A) 'What we now KNOW for certain' — groups add new confirmed facts with blue sticky notes (e.g., 'Water is densest at 4 °C'; 'Volume expansion: $\Delta V = \gamma V_0 \Delta T$ '; 'Ice has a hexagonal lattice with greater spacing than liquid water'). (B) 'What EXPLAINS the crack' — groups add one causal chain arrow card (pre-printed template): 'Water cools below 4 °C → expands → pressure builds → exceeds tensile strength of glass → crack.' (C) 'What we are STILL wondering' — groups add any residual questions raised by the experiment (e.g., 'Why does the hexagonal lattice have bigger spacing?'; 'Does salt water also expand when frozen?'; 'What happens to Lake Victoria fish in cold seasons?'). (D) 'How this connects to LESSONS 1–3' — groups draw a connecting thread on the board linking today's volumetric expansion ($\gamma \approx 3\alpha$) back	Say: 'Our DQB is a living record of our growing understanding. Lessons 1 to 3 gave us the tools — α , linear expansion, thermal stress. Today's experiment gave us the key evidence. Now we update the board.' Assign groups to zones explicitly. Give 5 minutes for group discussion before writing sticky notes. Circulate and prompt Zone C groups: 'Think beyond the bottle — where else in Kenya does water freezing and expanding matter? Think of pipes in high-altitude areas like Nyandarua or the Aberdares.' [WAIT 10 seconds.] After groups post, conduct a 4-minute whole-class DQB gallery walk — each group reads one new sticky note from a different group's zone and gives a thumbs-up if it is accurate or a side-tap on the shoulder of the poster group if they want to challenge it. Teacher closes by pointing to the causal chain in Zone B: 'This chain — right here — is the answer to our driving question. We are not done yet: Lesson 5 will test whether other liquids behave the same way, and Lesson 6 will bring it all together. But today, you cracked the mystery — pun intended.'	Driving Question Board — Cumulative Knowledge Tracking: The DQB functions as a public, evolving 'knowledge map' that externalises the class's collective understanding. Assigning groups to distinct zones prevents passive copying and ensures every learner actively generates new content. The causal-chain arrow card in Zone B is particularly powerful — it forces learners to express mechanism, not just correlation, linking directly back to the anchoring phenomenon. The 'connecting thread' in Zone D reinforces that science is cumulative: today's $\gamma \approx 3\alpha$ did not appear from nowhere but grew from Lesson 2's α .	Observable indicators: (1) Zone A sticky notes must contain at least one piece of quantitative knowledge (a formula, a temperature value, or a calculated result) — qualitative-only notes prompt teacher follow-up. (2) Zone B causal chain must include all four links (cooling → expansion → pressure → crack) — missing links are flagged during gallery walk. (3) Zone C questions are assessed for scientific curiosity and connection to lesson content — irrelevant questions indicate the learner has not yet connected experimental evidence to the driving question. Teacher records which groups populated each zone and cross-references with CER frames collected in Phase 3.

	to the linear expansion (α) from Lesson 2. The teacher photographs the updated DQB for the ARES platform record.			
Model Building Phase  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their cumulative class model (started in Lesson 1 as a simple particle diagram of the bottle before freezing). Today's revision task has two components: (A) MOLECULAR SCALE — Each learner adds a new panel to their model notebook showing: (i) liquid water molecules at 20 °C (random arrangement, moderate spacing, label: 'Volume = V_0'); (ii) water molecules at 4 °C (closest packing, label: 'Maximum density, minimum volume'); (iii) ice crystal at 0 °C (hexagonal lattice, clearly larger intermolecular spacing than panel ii, label: 'Volume = $V_0 + \Delta V$; $\Delta V = \gamma V_0 \Delta T$'). Arrows between panels show the direction of volume change. (B) MACROSCOPIC SCALE — Learners add quantitative annotations to their bottle diagram: the calculated ΔV for 500 ml of water freezing, an arrow showing outward pressure direction on the glass wall, and a label indicating 'Pressure exceeds tensile strength of glass → crack'. Groups then do a 3-minute 'Model Critique' — they swap notebooks with another group and use a printed 'Model Quality Checklist' (Does it show molecular spacing increasing? Does it include a formula? Does it connect micro to macro? Does it explain the crack?) to give two stars and one wish. Learners revise their models based on peer feedback before submitting notebooks.	Distribute the Model Quality Checklist. Say: 'Your model is not a drawing — it is an argument. Every arrow, every label must be doing explanatory work. If you can remove a label and the model still makes sense, the label was decorative, not scientific.' [Give 10 minutes for individual model revision.] Circulate and probe: 'Show me in your model exactly where the pressure builds. Now show me where the glass fails. Can I see that in your diagram?' [WAIT 12 seconds between question and follow-up.] Cold-call 2 learners to display their models under the document camera (or hold up for class): 'Class, using the checklist — two stars and one wish for this model. Hands up.' [Take 3–4 contributions per model shown.] After peer critique swap: 'You have 3 minutes to implement at least one wish from your partner's feedback — make the revision in a different colour so we can see the growth.' Close the lesson: 'From a single cracked bottle, we have now built a model that works at the molecular scale and the macroscopic scale. That is what physicists do — they connect the invisible to the visible. Well done.' Remind learners: 'For Lesson 5, we will test other liquids and gases — bring a curious question from your DQB.'	Cumulative Model Revision + Peer Critique: The multi-panel molecular model operationalises the NGSS practice 'Developing and Using Models' — learners must represent the same phenomenon at two scales (molecular and macroscopic) and connect them explicitly. This dual-scale requirement prevents learners from holding disconnected micro and macro explanations. The Model Quality Checklist scaffolds peer critique with specific, scientific criteria rather than aesthetic judgement, teaching learners to evaluate models the way scientists do — by their explanatory power. Revision in a contrasting colour makes learning progress visible and builds metacognitive awareness.	Observable indicators: (1) Molecular panel iii must show clearly larger intermolecular spacing than panel ii — not just labelled but visually represented. (2) The formula $\Delta V = \gamma V_0 \Delta T$ must appear with values substituted (not just written abstractly). (3) The macroscopic bottle diagram must include a directional pressure arrow and an explicit link to the crack. (4) Model Quality Checklist peer scores are collected — any model scoring fewer than 3 out of 4 checklist criteria is flagged for teacher follow-up in Lesson 5. Teacher also checks: do learners who held the 'water always shrinks' misconception in Phase 1 now show the correct anomalous behaviour in their molecular panels? This is the key conceptual shift indicator for the lesson.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) **CONCEPTUAL SHIFT** — Compare the Phase 1 sticky-note tally (AGREE/DISAGREE/UNSURE about the bottle cracking) with an identical tally administered at the end of Phase 3. What percentage of learners shifted from DISAGREE or UNSURE to AGREE with correct CER reasoning? If fewer than 70% shifted, what was the sticking point — the molecular explanation, the formula, or the graph interpretation? (2) **EXPERIMENTAL DATA QUALITY** — Did all groups obtain data clearly showing the volume minimum near 4 °C? If not, was this due to equipment limitations (thermometer resolution, capillary tube diameter), procedural errors (flask not fully sealed, ice bath not cold enough), or conceptual errors (reading the capillary incorrectly)? What adjustments will you make to the apparatus setup for next year's cohort? (3) **GENDER AND PARTICIPATION** — Were girls as likely as boys to be assigned the thermometer-reading and data-recording roles, or did gendered patterns emerge? Intentionally rotate roles in Lesson 5 to ensure equitable participation in all hands-on tasks. (4) **MATHEMATICAL DEMAND** — Did learners successfully navigate the derivation of $\beta \approx 2\alpha$ and $\gamma \approx 3\alpha$, or did the binomial approximation step cause confusion? If many groups struggled, consider adding a visual 'area of the border strip' diagram to the Derivation Card in future iterations. (5) **DQB VITALITY** — Are the questions in Zone C (What we are still wondering) scientifically generative — pointing toward Lessons 5 and 6? If learners' questions were superficial, model a 'rich question' before the next DQB update: e.g., 'Does the anomalous expansion of water affect the density of Lake Victoria at different depths — and could that affect the fish populations that Kenyan fishing communities depend on?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners set up a flask-and-capillary-tube apparatus, cooled water from room temperature toward 0 °C while measuring the capillary column height at regular temperature intervals, and plotted a temperature–volume graph. They observed that the capillary column — and therefore the water volume — decreased as temperature dropped from 20 °C to 4 °C, then increased again as temperature dropped further from 4 °C to 0 °C, forming a distinctive U-shaped curve with a minimum at approximately 4 °C.
What did I learn?	Learners learned that: (1) superficial expansivity $\beta \approx 2\alpha$ and volumetric expansivity $\gamma \approx 3\alpha$ can be derived mathematically from linear expansivity α , and used in the formula $\Delta V = \gamma V_0 \Delta T$; (2) most substances contract when cooled, but water is anomalous — it expands below 4 °C because water molecules arrange into a hexagonal ice crystal lattice with greater intermolecular spacing than liquid water; (3) water reaches its maximum density at 4 °C; (4) this anomalous expansion is the direct cause of the cracked frozen bottle in the anchoring phenomenon, as the expanding ice generated pressure exceeding the tensile strength of the glass.
How does this explain the phenomenon?	The sealed glass bottle cracked because water, unlike most substances, expands when it freezes. As the water cooled below 4 °C, its molecules arranged into a hexagonal ice crystal lattice in which the intermolecular spacing is greater than in liquid water — so the same mass of water occupies a larger volume as ice. Confined inside a sealed bottle with no room to expand, the growing ice exerted outward pressure on the glass walls. When this pressure exceeded the tensile strength of the glass, the bottle cracked. The volumetric expansion can be quantified using $\Delta V = \gamma V_0 \Delta T$, and the temperature–volume graph of water provides direct experimental evidence of this anomalous behaviour.

LESSON 5 (40 minutes): Applications of Thermal Expansion: Engineering Solutions

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate and evaluate how engineers and technologists apply knowledge of thermal expansion to design safe, functional structures and devices in real Kenyan contexts
Knowledge	Learners will be able to: (1) identify at least four real-world applications of thermal expansion in Kenyan engineering and everyday technology; (2) explain how the linear expansion formula $\Delta L = \alpha L_0 \Delta T$ is used to calculate required expansion gap sizes in structures; (3) describe how bimetallic strips exploit differential expansion to function as thermostats
Skills	Learners will be able to: (1) apply the linear expansion formula to solve a structured design problem involving a steel bridge beam in Kisumu; (2) evaluate engineering solutions by comparing calculated expansion values with design specifications; (3) present and justify a proposed design solution using scientific reasoning linked to expansion data
Attitudes	Learners will appreciate that physics knowledge directly underpins engineering decisions that affect community safety — from roads Kenyans drive on to irons used in homes — and develop confidence in applying formulae to real design challenges
Key Inquiry Question	How is the expansion of solids, liquids, and gases applied in everyday life?
Purpose in Storyline	Having established in Lessons 2 to 4 that materials expand predictably with temperature and that water expands anomalously below 4 degrees C causing the cracked bottle, Lesson 5 now asks: how do engineers harness or accommodate this inevitable expansion? Learners move from explaining the phenomenon to evaluating solutions, building the engineering design dimension of their final explanation poster in Lesson 6.
Safety Notes	No hazardous materials are used in this lesson. If a physical bimetallic strip or electric iron thermostat is demonstrated, the teacher should ensure the iron is handled only by the teacher on a heat-safe mat and that no learner touches the heated surface. All calculations involve pencil, paper, and provided data cards only.

B. LESSON OVERVIEW

Learners open the lesson by revisiting the cracked bottle phenomenon and asking: if engineers know materials expand, how do they stop structures from cracking the way the bottle did? Working in groups of four, learners receive Kenya-context application cards describing expansion joints on the Thika Superhighway, bimetallic strip thermostats in electric irons, fitting of metal tyres on wooden cart wheels, and riveting in metal roofing sheets. Each group analyses one application, identifies the expansion principle at work, and presents a two-minute explanation to the class. The teacher then leads a whole-class worked example calculating the required expansion gap for a 10 m steel bridge beam in Kisumu exposed to a 35 degree C seasonal range, using $\Delta L = \alpha L_0 \Delta T$ with α for steel = 12×10^{-6} per degree C. Learners then independently solve a parallel problem involving overhead power line cables run by Kenya Power between Nairobi and Thika. The lesson closes with learners updating the Driving Question Board to record: engineers use the same expansion formula that explains the cracked bottle to deliberately design gaps and devices, turning a destructive force into a controlled engineering tool.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: The history of atomic chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners re-examine the projected image of the cracked frozen bottle from Lesson 1. The teacher poses the question: the bottle cracked because water expanded and engineers could not control it — but what if an engineer knew in advance that a material would expand? What would they do to stop cracking or breaking? Learners individually write a one-sentence prediction in their notebooks: how might an engineer use the expansion formula to prevent damage to a long steel structure? They share predictions with a shoulder partner before the teacher takes two or three oral responses.	Display the cracked bottle image and say: We have spent four lessons understanding why this bottle cracked. Today we flip the question — if you were designing a steel bridge in Kisumu and you knew the metal would expand every hot season, what would you actually build into the design to stop the bridge from cracking like this bottle? Give learners 60 seconds of silent writing time, then say: Share your idea with your partner. You have 30 seconds each. After paired sharing, cold-call two learners and record their key words on the board — likely words will include gap, space, joint, or room to expand. Do not evaluate — say: Let us test these ideas against what engineers actually do.	Anchoring phenomenon re-connection: the cracked bottle is used as a negative engineering outcome that motivates the search for positive design solutions. Learners are positioned as engineers rather than just observers, activating prior knowledge of $\Delta L = \alpha L_0 \Delta T$ in a design frame.	Listen for whether learners spontaneously reference the expansion formula or expansion coefficient from Lesson 3. If no learner mentions the formula, ask: which formula from last lesson could an engineer use to calculate exactly how much space to leave? This diagnostic reveals whether formula knowledge is transferable to new contexts.
Observe Phase  VIDEO: The history of atomic chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of four receive one Application Card each. The four cards are: Card A — Expansion Joints on the Thika Superhighway: photographs and a short text describing the rubber-filled gaps cut into the concrete road surface every 6 m, explaining that concrete has a linear expansivity of approximately 10×10^{-6} to the power of minus 6 per degree C and Nairobi temperatures range from 10 degrees C to 30 degrees C. Card B — Bimetallic Strip Thermostat in an Electric Iron: a labelled diagram showing a strip of brass bonded to a strip of invar steel, with brass having a higher expansivity so the strip bends toward the invar side on heating, breaking an electrical circuit and switching the iron off. Card C — Fitting a Metal Tyre onto a Wooden Cart Wheel in Rural Kenya: a description of	Distribute cards and say: You have 8 minutes. Your group must answer three questions about your application: one — what type of expansion is involved? Two — is the engineer accommodating expansion, exploiting it, or preventing it? Three — how does this connect to the cracked bottle? Circulate to each group. For Card B groups struggling with the bimetallic strip, hold up a physical bimetallic strip (or a diagram) and say: Look — two metals are bonded together. Brass expands more than invar for the same temperature rise. What will the strip do when heated? Wait 10 seconds. For Card C groups, ask: Why does the tyre grip tighter when it cools rather than when it is hot? Wait 10 seconds before offering: think about what happens to the diameter of the tyre as it cools. At the 8-minute mark, bring	Jigsaw-style card analysis ensures every learner encounters all four applications through peer teaching. The three-question scaffold (type of expansion, engineering approach, connection to phenomenon) structures observation toward scientific reasoning rather than surface description. The physical or diagrammatic bimetallic strip provides a concrete referent for an otherwise abstract device.	During circulations, listen for correct identification of expansion type and for unprompted use of the terms linear expansivity or expansion coefficient. After each group presentation, use the question: how does this connect to the cracked bottle? as a probe — learners who can articulate that both involve volumetric or linear expansion of a solid under temperature change demonstrate conceptual transfer. Note any group that conflates thermal contraction with cooling-only scenarios.

	blacksmiths heating a slightly undersized iron tyre until it expands enough to slip over the wooden wheel, then cooling it with water so it contracts and grips tightly. Card D — Riveting Corrugated Metal Roofing Sheets: a description of how roofers in Kenyan construction leave small gaps between adjacent sheets and use dome-headed rivets that allow slight movement, preventing buckling during afternoon heat. Each group reads their card, identifies the specific type of expansion (linear, superficial, or volumetric), names the engineering principle being exploited (accommodate, exploit, or prevent expansion), and prepares a two-minute verbal explanation with one key diagram sketch.	the class together for group presentations. After each presentation say: Does anyone want to add or challenge anything in that explanation? Allow 30 seconds of open response.		
Explain Phase  VIDEO: The history of atomic chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher leads a whole-class worked example on the board. Problem: A steel bridge beam in Kisumu is 10 m long at an installation temperature of 20 degrees C. The seasonal temperature in Kisumu ranges from 18 degrees C to 35 degrees C, giving a maximum temperature rise of 15 degrees C. The linear expansivity of steel is 12×10 to the power of minus 6 per degree C. What minimum expansion gap must the engineer leave at each end of the beam? Learners copy the problem into their notebooks and attempt the calculation individually for 3 minutes before the teacher works through it step by step on the board: $\Delta L = \alpha L_0 \Delta T = (12 \times 10 \text{ to the power of minus 6}) \times 10 \times 15 = 1.8 \times 10 \text{ to the power of minus 3 m} = 1.8 \text{ mm}$. The teacher then	Before writing the bridge problem, say: Before I give you the numbers, tell me — what formula will we use and what does each symbol mean? Cold-call one learner. Then write the problem on the board and say: Work this out alone in silence for 3 minutes. I want to see your formula written out, your substitution, and your answer with units. Start now. After 3 minutes say: Keep your book open to your attempt. Watch how I do it and put a tick where you got it right or an arrow where your working differed. Work through the solution narrating each step aloud. After reaching the answer say: 1.8 mm might sound tiny — but what happened to the glass bottle when the ice had nowhere to go? Pause for 5 seconds of wait time. Take one response, then say: Exactly — even small expansions generate	The worked example uses gradual release: learners attempt first, observe teacher model second, then self-assess against the model. The parallel independent problem immediately transfers the skill to a different material and context. Explicit connection back to the cracked bottle at the explanation stage maintains the anchoring phenomenon as the explanatory touchstone throughout. Narrating each step of the formula aloud models scientific communication.	Circulate during the 5-minute independent problem and look at three specific checkpoints in learner books: (1) Is the formula written symbolically before numbers are substituted? (2) Is ΔT calculated correctly as 40 minus 15 = 25 degrees C and not confused with the final temperature? (3) Does the final sentence demonstrate causal reasoning — that the cable sags to allow room for thermal expansion? Record the names of learners who make the ΔT error for targeted support in Lesson 6.

	connects: the engineer must leave at least 1.8 mm at each end, so a total of 3.6 mm across the full beam, otherwise the beam pushes against its supports with the same kind of force that cracked the glass bottle. Learners then independently solve a parallel problem: A Kenya Power overhead copper conductor cable is 200 m long at 15 degrees C. The temperature rises to 40 degrees C on a hot Nairobi afternoon. The linear expansivity of copper is 17×10^{-6} per degree C. Calculate the increase in length and explain why engineers design the cable to sag rather than stretch taut between pylons. After 5 minutes of independent work, learners compare answers with a partner. The expected answer is $\Delta L = 17 \times 10^{-6} \times 200 \times 25 = 0.085 \text{ m} = 8.5 \text{ cm}$.	enormous forces when contained. That is the engineering lesson from our cracked bottle. For the independent copper cable problem, say: This time you are the engineer. After solving, write one sentence explaining why the cable sags. After partner comparison, ask: Who got 8.5 cm? Show of hands. If fewer than two-thirds raise hands, rework the substitution step on the board.		
Driving Question Board (DQB) Creation  VIDEO: The history of atomic chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board established in Lesson 1. The teacher reads aloud two outstanding questions from the board that this lesson addresses: How do engineers stop structures from expanding and cracking like the bottle? and Are there times when expansion is useful rather than destructive? Learners write one new sticky note each. The note must complete the sentence stem: Engineers use the expansion formula to _____. Sample learner responses include: leave gaps in bridges so beams do not push and crack, make bimetallic strips bend and switch off irons, and shrink metal tyres onto wheels. Learners post their notes on the DQB under a new column	Point to the DQB and say: In Lesson 1 you asked whether expansion could be useful. Look at what we have discovered today — point to the application card examples still displayed. Now write a sticky note that begins with the words: Engineers use the expansion formula to _____. You have 2 minutes. While learners write, prepare a fresh column on the DQB labelled Engineering Solutions. After posting, read two or three notes aloud and say: Notice something — every one of these notes uses the same formula we first met in Lesson 3 when we heated a metal rod over a Bunsen burner. That formula connects the cracked bottle, the bridge in Kisumu, and the iron in your home. Ask: Is there any	The DQB update makes the progression of the learning sequence visible to learners — they can physically see Lesson 1 questions being resolved by Lesson 5 evidence. The sentence stem scaffolds abstract formula knowledge into applied declarative language. The teacher note models the big-picture synthesis move that learners will need to make independently in Lesson 6.	Read all posted sticky notes during or after the lesson. Flag any note that only names an application without connecting it to the formula or to the phenomenon — these learners need additional scaffolding in Lesson 6 when producing their explanation poster. Look specifically for learners who can articulate directionality: that expansion without accommodation produces destructive force, whereas designed gaps or exploited bending convert that force into function.

	labelled Engineering Solutions. The teacher adds one teacher note reading: The same force that cracked the bottle is harnessed in thermostats, tyre fitting, and road joints — expansion is not always the enemy.	question left on our board that we still have not answered? Allow 30 seconds of silent reading of the board. Take one or two nominations and say: We will return to those in our final lesson when we build our complete explanation.		
Model Building Phase  VIDEO: The history of atomic chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their cumulative particle model notebooks — the model they have been building since Lesson 1. They add a new panel labelled Lesson 5: Engineering Response to Expansion. In this panel they draw a simple cross-section of a bridge beam showing: (1) the beam at low temperature with particles close together and the beam fitting snugly between two supports with a visible gap at each end; (2) the same beam at high temperature with particles vibrating at greater amplitude, the beam elongated, and the gap now partially closed but not yet in contact with the supports. Underneath the diagram they write the equation they used today and one sentence explaining the engineering principle. Learners who finish early add a third sub-panel showing a bimetallic strip bending on heating and labelling which metal has the higher expansivity. At the close of the lesson the teacher calls on one learner to share their diagram under the document camera or by holding it up, and the class offers one thing the diagram shows well and one thing that could be added.	Say: Open your cumulative model notebook to a fresh page. Label it Lesson 5: Engineering Response to Expansion. Draw the bridge beam scenario in two stages — cold and hot. Show me the particles, the beam length, and the gap. You have 4 minutes. Circulate and look for common error: learners drawing particles as if they move apart (increasing spacing) rather than vibrate more vigorously in place. If seen, say quietly: Remember from Lesson 3 — the particles do not fly apart. What changes is how hard they vibrate. Does that change the length of the beam? Wait 5 seconds. When selecting the sharing learner, choose one whose diagram correctly shows vibration amplitude rather than inter-particle spacing change, and narrate why this is accurate: This diagram shows particles staying bonded but vibrating more — that increased vibration pushes the ends of the beam outward. That is exactly why we need the gap. Before closing say: In our final lesson we will put every panel of this model together to write our full explanation of the cracked bottle. Make sure all five panels are complete.	The cumulative particle model is the through-line artefact of the entire unit. Adding the engineering panel in Lesson 5 ensures learners do not treat applications as separate from particle-level explanation — the model explicitly represents both scales simultaneously. Peer gallery feedback using the one strength and one addition protocol builds collaborative scientific communication norms ahead of the Lesson 6 synthesis poster.	Collect or photograph three to four particle model notebooks before learners leave. Check specifically: (1) Is the gap drawn at both ends of the beam (not just one)? (2) Are particles shown vibrating (wavy arrows or blurred positions) rather than simply spreading apart? (3) Is $\Delta L = \alpha L_0 \Delta T$ written with correct symbol usage? Use these as the entry point for Lesson 6 discussion by projecting an anonymised strong example and asking the class to identify what makes it a good scientific model.

D. TEACHER REFLECTION

After the lesson, consider: Did learners spontaneously invoke the expansion formula when analysing the application cards, or did they describe applications in purely descriptive terms without mathematical connection? If the latter, the parallel problem in Lesson 6 should be preceded by an additional guided example. Did the DQB sticky notes show causal language (the beam expands so the engineer leaves a gap) or only labelling (expansion joints)? Causal language indicates readiness for the Lesson 6 synthesis poster. Were learners able to distinguish between accommodating expansion (bridge gaps, cable sag) and exploiting expansion (bimetallic strip, tyre fitting)? This distinction is important for the poster because it shows that the same physical principle can be both a hazard and a resource depending on design intent. Note which learners made the ΔT substitution error in the copper cable problem and ensure they receive targeted support at the start of Lesson 6 before attempting the cracked-bottle volumetric calculation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed four real Kenyan applications of thermal expansion through application cards: expansion joints on the Thika Superhighway, bimetallic strip thermostats in electric irons, metal tyre fitting on wooden cart wheels, and riveted corrugated roofing. They also observed a worked calculation showing that a 10 m steel beam in Kisumu requires at least 1.8 mm of expansion gap per end to safely accommodate a 15 degree C seasonal temperature rise.
What did I learn?	Learners learned that engineers use the linear expansion formula $\Delta L = \alpha L_0 \Delta T$ to calculate the exact gap size required in structures such as bridges and roads, preventing destructive buckling or cracking. They learned that bimetallic strips exploit differential expansion of two bonded metals to create a temperature-controlled switch, and that thermal expansion can be deliberately engineered to be useful (tyre fitting, thermostat) rather than only destructive (cracked bottle). They practised applying the formula to a new context involving Kenya Power copper cables expanding by 8.5 cm over 200 m on a hot Nairobi afternoon.
How does this explain the phenomenon?	Learners explained that the same physical principle that caused the bottle to crack — materials expanding when heated and generating force when confined — is the principle engineers consciously calculate and design around in every structure exposed to temperature changes in Kenya. The cracked bottle represents an uncontrolled application of thermal expansion; Kenyan engineering infrastructure represents the controlled and calculated application of the same phenomenon, making the bottle not just a curiosity but a model for understanding why expansion joints, sagging cables, and bimetallic switches exist.

LESSON 6 (80 minutes): Model Revision, Synthesis, and DQB Resolution — Explaining the Crack

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate and synthesise all learning from the unit by returning to the anchoring phenomenon with full explanatory power, revising the cumulative particle model, resolving all outstanding Driving Question Board questions, and producing a final annotated explanation that connects the cracked bottle to temperature scales, thermal expansion coefficients, anomalous water behaviour, and real-world engineering design.
Knowledge	Learners will demonstrate understanding that: (1) water expands anomalously upon freezing because hydrogen bonding arranges water molecules into a hexagonal lattice with greater intermolecular spacing than liquid water; (2) the volumetric expansion of water ($\Delta V = \beta V_0 \Delta T$, with $\beta \approx 9 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ for the freezing transition) generates pressures sufficient to fracture glass; (3) solids, liquids, and gases each expand at characteristically different rates captured by their respective expansion coefficients; (4) temperature scales (Celsius, Kelvin, Fahrenheit) provide consistent frameworks for quantifying thermal change; and (5) engineers account for thermal expansion in pipelines, bridges, railway tracks, and container design across Kenya and globally.
Skills	Learners will be able to: (a) construct a fully annotated, multi-scale particle model that traces the cracked-bottle phenomenon from molecular motion to macroscopic crack patterns; (b) apply the linear and volumetric expansion formulae to solve quantitative problems; (c) synthesise evidence from all six lessons to produce a coherent, written final explanation; (d) evaluate the accuracy and completeness of their Lesson 1 prediction model against their final model; (e) collaborate to resolve residual questions on the Driving Question Board with evidence-based reasoning.
Attitudes	Learners will appreciate that scientific understanding is iterative — initial models are refined as evidence accumulates — and will value precision, intellectual honesty, and collaborative sense-making as core scientific dispositions. They will recognise that anomalous phenomena (water expanding on cooling) are not errors but windows into deeper scientific truth.
Key Inquiry Question	Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?
Purpose in Storyline	This is the capstone lesson of the unit. Learners arrive with all the conceptual tools built across Lessons 1–5: temperature scales, particle kinetic theory, linear expansion of solids, volumetric expansion of liquids and gases, anomalous expansion of water, and expansion coefficients. Lesson 6 is the moment of full explanation — learners return to the cracked bottle, revise their cumulative particle models one final time, close every open question on the Driving Question Board, and demonstrate mastery by producing an annotated explanation poster that a non-scientist could understand. The lesson honours the storyline arc: from puzzlement (Lesson 1) to complete, quantitative, model-based explanation (Lesson 6).
Safety Notes	No new hazardous materials are introduced in this lesson. If the pre-cracked demonstration bottle from Lesson 1 is handled again, learners must not touch broken glass edges directly — use tongs or a cloth. All sharp glass fragments should be contained. Remind learners that freezing pressurised sealed containers at home is dangerous and should never be attempted without engineering-grade containers.

B. LESSON OVERVIEW

In this 80-minute capstone lesson, learners return to the anchoring phenomenon — the cracked frozen glass bottle — with the full conceptual and quantitative toolkit built across the unit. The lesson opens with a rapid 'What We Now Know' gallery walk that surfaces all accumulated evidence. Learners then engage in a structured Final

Explanation writing task, producing an annotated explanation poster that links the cracked bottle to particle behaviour, temperature scales, volumetric expansion formula, anomalous water behaviour, and engineering applications. A paired model-comparison activity contrasts each learner's Lesson 1 naive model with their final revised particle model, making growth in understanding visible. The Driving Question Board is formally closed as groups resolve every outstanding sticky-note question with evidence from the unit. The lesson concludes with a whole-class 'explain the crack' cold-call synthesis, teacher-led DQB celebration, and a forward-connection to real Kenyan engineering contexts (water pipeline design in Nairobi, railway expansion gaps on the SGR line, storage tank design at the Port of Mombasa). NGSS Science and Engineering Practices embedded: Constructing Explanations, Developing and Using Models, Engaging in Argument from Evidence, and Obtaining/Evaluating/Communicating Information.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Learners individually retrieve their Lesson 1 prediction cards (stored in their science notebooks or distributed by the teacher). They silently re-read their original answer to: 'Why do you think the bottle cracked?' and their original particle diagram. They then write a one-sentence 'Then vs. Now' statement in their notebooks — e.g., 'In Lesson 1 I thought the cold air pushed in and cracked the glass, but now I know that water molecules form a hexagonal lattice with greater spacing when they freeze, increasing volume by about 9% and generating outward pressure.' Learners share with a shoulder partner for 2 minutes. Selected pairs are cold-called to share their 'Then vs. Now' contrasts with the class. The teacher records 4–5 contrasting pairs on the board in a two-column T-chart: LESSON 1 MODEL FINAL MODEL. This phase activates metacognitive awareness of conceptual growth and anchors the lesson back to the phenomenon.	Distribute Lesson 1 prediction cards (pre-retrieved). Say: 'Before we do anything else today, I want you to travel back in time to Day 1. Read your Lesson 1 prediction silently. Do not change anything — honour what you thought then.' [Allow 2 minutes silent reading.] Then say: 'Now write one sentence in your notebook — start with: In Lesson 1 I thought... but now I know...' [Allow 3 minutes. WAIT TIME: do not speak for the full 3 minutes.] After writing: 'Turn to your shoulder partner. Share your Then vs. Now. You have 2 minutes — each person must speak.' [Set a 2-minute timer.] Cold-call 4 pairs. Ask each pair: 'What is the single biggest shift in your thinking?' Record responses in the T-chart. Then point to the cracked bottle displayed at the front and say: 'This bottle cracked because of something your Lesson 1 self could not yet fully explain. By the end of today, every single one of you will be able to explain it completely — in words, in a particle diagram, and with a formula. Let's go.'	Strategy: Metacognitive Retrieval and Contrastive Pairs. By explicitly retrieving and contrasting their naive model with their current model, learners consolidate conceptual change and make their own learning trajectory visible. This is not review — it is evidence of understanding. The T-chart on the board externalises the class's collective growth and creates a shared reference point for the synthesis work ahead.	Observable indicator: Every learner produces a written 'Then vs. Now' sentence within 3 minutes. Teacher circulates and scans for misconceptions that persist (e.g., learners who still cite 'cold air pressure' as the cause). Flag these learners for targeted support during Phase 3. Cold-call responses reveal the range of conceptual depth across the class — look for learners who reference particle spacing, hydrogen bonding, volumetric expansion, and the formula $\Delta V = \beta V_0 \Delta T$ as markers of high-level understanding.
Observe Phase	Six evidence stations are arranged around the classroom walls, each	Before the gallery walk: 'You have 6 stations. Each one is a piece of	Strategy: Evidence Carousel with Connection Prompts. The gallery	Observable indicator: Each learner's notebook contains 6

VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Thermal Energy (Flexbook) Source: CK-12 Search ARES for similar readings	summarising the key finding from one lesson in the unit. Each station contains: a headline statement, a key diagram or data table, one quantitative example, and a 'Connection to the Cracked Bottle' prompt. Station contents: (1) Temperature Scales — Celsius, Kelvin, Fahrenheit conversion diagram; water freezes at 0°C/273 K; (2) Particle Kinetic Theory — diagram of liquid water molecules vs. hexagonal ice lattice, showing greater spacing; (3) Linear Expansion of Solids — formula $\Delta L = \alpha L_0 \Delta T$, worked example with steel railway track on the SGR Nairobi–Mombasa line; (4) Volumetric Expansion of Liquids — formula $\Delta V = \beta V_0 \Delta T$, worked example with 500 ml water bottle; (5) Anomalous Expansion of Water — density vs. temperature graph showing maximum density at 4°C; (6) Gases and Real-World Applications — Charles's Law, expansion gaps in Nairobi's Thika Superhighway, pipeline design at the Mombasa refinery. Learners move in groups of 3–4, spending 4 minutes at each station. Each learner records: (a) one key piece of evidence from the station, (b) how it connects to the cracked bottle. Groups have 2 minutes to travel between stations. Total gallery walk: approximately 24 minutes.	evidence your class has built over the last five lessons. Your job is not just to read — it is to collect and connect. At every station, write in your notebook: What is the key evidence? AND How does this connect to the cracked bottle?' Assign groups of 3–4 by numbering off. Say: 'In your groups, one person is the Recorder, one is the Timekeeper, one is the Reporter. Rotate these roles at every station.' [Ring a bell or clap to signal station transitions every 4 minutes.] Circulate during the gallery walk. At Station 4 (volumetric expansion), pause groups and ask: 'If β for water near freezing is approximately 9×10^{-5} per degree Celsius, and our bottle holds 500 ml, and the temperature drops from 4°C to 0°C, what volume change does the formula predict?' [WAIT TIME: 15 seconds.] Cold-call one recorder to share their group's calculation. At Station 5 (anomalous expansion), ask: 'What makes water different from almost every other substance on Earth? Point to the evidence on this poster.' [WAIT TIME: 10 seconds.] After the gallery walk, bring the class together: 'Reporters — in 30 seconds each, tell me the one piece of evidence from any station that most surprised you or that most powerfully explains the crack.' Cold-call 5 reporters.	walk is not passive reading — the 'connection to the cracked bottle' prompt at every station forces learners to perform transfer: applying each concept to the anchoring phenomenon. By physically moving through stations, learners experience the evidence as a coherent body rather than isolated facts. The role assignments (Recorder, Timekeeper, Reporter) ensure equitable participation within groups.	completed evidence entries (one per station) with explicit connection statements. Teacher scans notebooks during circulation — look for learners who can articulate the volumetric expansion formula, identify the anomalous density-temperature relationship, and connect both to the cracked bottle. Reporters' 30-second summaries reveal depth of synthesis. Red flag: any learner who cannot articulate why water is anomalous — pull these learners into a teacher-facilitated small group during Phase 3.
Explain Phase VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum) Search ARES	This is the heart of the lesson. Learners work in pairs to construct a Final Explanation Poster on A3 paper (or a folded A2 sheet). The poster must contain five labelled sections: (A) The Phenomenon — a sketch of the cracked bottle with labels identifying the crack pattern, the ice plug, and the direction of	Distribute A3 paper and coloured markers. Say: 'This is your final explanation. Imagine you are sending this poster to a Form 2 student who missed all six lessons. Your poster must explain the cracked bottle so completely that they never need to ask another question.' Display the five	Strategy: Constructing Explanations with Multi-Representational Output (NGSS SEP 6). Requiring learners to represent the phenomenon simultaneously in words, diagrams, mathematics, and real-world application forces deep integration of knowledge. The	Observable indicator: Each poster contains all five labelled sections, a correct and annotated particle diagram, a correct quantitative calculation ($\Delta V \approx 0.18$ ml), a density-temperature graph with the 4°C peak, and at least two Kenya-specific engineering examples. Teacher evaluates during

for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	outward pressure. (B) The Particle-Level Explanation — a before/after particle diagram showing liquid water molecules (close-packed, disordered) vs. ice molecules (hexagonal lattice, greater spacing), with annotations explaining why spacing increases (hydrogen bonding, tetrahedral coordination). (C) The Quantitative Explanation — a worked example using $\Delta V = \beta V_0 \Delta T$ to calculate the volume change of 500 ml of water cooling from 4°C to 0°C ($\beta = 9 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$), showing that $\Delta V \approx 0.18$ ml, and explaining that because the glass bottle is rigid and sealed, this expansion generates enormous internal pressure sufficient to crack glass (tensile strength of glass $\approx 33\text{--}100$ MPa). (D) Why Water is Anomalous — a mini density-vs-temperature graph showing the 4°C maximum density peak, with an explanation of why most substances contract on cooling but water does not. (E) Engineering Implications — two real Kenyan examples: (i) why water supply pipes in cold highland areas of Mount Kenya or Kericho sometimes burst in cold seasons, and (ii) how engineers design expansion joints in the SGR railway track between Nairobi and Mombasa to accommodate linear thermal expansion. Pairs display their posters and participate in a 10-minute 'Poster Parade' where they give and receive peer feedback using the sentence stems: 'Your explanation of ____ is strong because ____ / I suggest you strengthen ____ by adding ____.'	required sections on the board. Say: 'You have 20 minutes. Both partners must write — I will be checking that both hands are moving.' [Set a 20-minute timer.] Circulate actively. At the 7-minute mark, check that ALL pairs have completed sections A and B. If any pair is stuck on the particle diagram, pose: 'Think about what you saw at Station 2 in the gallery walk. What did the hexagonal lattice look like compared to the liquid? Draw that.' [WAIT TIME: 10 seconds.] At the 14-minute mark, check that section C is underway. If pairs are struggling with the calculation, write on the board: $\Delta V = \beta V_0 \Delta T \rightarrow \Delta V = (9 \times 10^{-5})(500)(4)$ and ask: 'What is 9 times 500 times 4, divided by 10 000? Take 15 seconds.' [WAIT TIME: 15 seconds.] Cold-call one pair: 'What did you get?' After posters are complete, announce the Poster Parade: 'Stand up. Take your poster. Walk to the pair to your left. Read their poster. Use the sentence stems on the board. You have 5 minutes, then rotate once more.' After the parade: 'Return to your seat. In 2 minutes, revise ONE section of your poster based on the feedback you received. Circle the revision in red so I can see your thinking grew.'	Poster Parade adds the critical dimension of peer argumentation (NGSS SEP 7): learners must defend their explanations and evaluate others', further consolidating understanding. The red-circle revision step makes metacognitive growth visible and honours the iterative nature of scientific modelling.	circulation using a rapid 3-point rubric per section: 3 = complete and accurate, 2 = partially correct or missing annotation, 1 = incomplete or misconception present. Peer feedback sentence stems generate observable written feedback that the teacher can scan after the lesson. The red-circle revision reveals responsiveness to feedback.
Driving Question	The Driving Question Board (maintained since Lesson 1) is	Bring out the DQB. Say: 'Look at this board. Every question on here	Strategy: Driving Question Board Closure with Evidence-Anchored	Observable indicator: Every DQB question receives either a green

Board (DQB) Creation  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	returned to the front of the room with all accumulated sticky-note questions visible. Learners count the total number of questions posted across all six lessons. Working in their groups of 3–4 from the gallery walk, each group is assigned 3–4 DQB questions to resolve. They must: (a) read the question aloud, (b) agree on a complete evidence-based answer using their notebooks, gallery walk notes, and final explanation poster, and (c) write the answer on a green sticky note and attach it beside the original question. Questions that cannot be fully answered are flagged with an orange sticky note marked 'Frontier Science — still being investigated.' The full class reconvenes: each group's reporter reads one question-and-answer pair aloud. The teacher leads the class in a visible 'closing ceremony' — physically moving all resolved questions to a RESOLVED column on the DQB. The driving question itself — 'Why did the sealed water bottle crack in the freezer?' — is the last question resolved, by the whole class together in a choral, structured response.	was asked by someone in this room — by you — because the cracked bottle genuinely puzzled you. Count the questions with me.' [Count aloud together.] 'Every one of these questions deserves a real answer. Today, we close the board.' Assign groups their questions: 'Group 1 takes the blue cluster on the top left — those questions about why water is different. Group 2 takes the yellow cluster — questions about the formula and the numbers. Group 3 takes the pink cluster — questions about what happens to other liquids in a freezer. Group 4 takes the green cluster — engineering and real-world questions.' [Allow 8 minutes for groups to draft answers on green sticky notes.] Circulate. If a group is debating an answer, do not resolve it for them — instead ask: 'What evidence from the gallery walk or your poster supports your answer? Point to it.' [WAIT TIME: 10 seconds.] During the whole-class resolution: cold-call each group's reporter. After each answer, ask the class: 'Do we agree? Thumbs up if yes, thumbs sideways if partly, thumbs down if you disagree.' Honour disagreements — ask: 'What evidence would change this answer?' For the driving question itself, say: 'The question that started everything. I want to hear it answered — all together — in three sentences. First sentence: the phenomenon. Second sentence: the particle-level reason. Third sentence: the formula. You have 2 minutes to write your three sentences, then we say them together.' [WAIT TIME: 2 minutes writing.] Lead a structured choral reading of the	Resolution (NGSS SEP 8 — Obtaining, Evaluating, and Communicating Information). The DQB has been a living artefact of the class's curiosity across all six lessons. Formally closing it is not administrative — it is a meaning-making ritual that signals the completion of a genuine scientific investigation. The distinction between 'resolved' (green) and 'frontier science' (orange) models authentic scientific practice: not all questions have settled answers, and that is intellectually honest. The three-sentence choral answer to the driving question creates a shared, memorable, precise class statement of understanding.	resolved note or an orange frontier note — no question is left unaddressed. Teacher reads all green notes to assess accuracy of peer-generated answers. The thumbs-up/sideways/down protocol after each group's answer provides instant class-wide formative data. The three-sentence driving question answer (written in notebooks before the choral reading) is the clearest individual formative assessment of the lesson — teacher collects these at the end for review.
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		three-sentence answer.		
Model Building Phase  VIDEO: Linear Perspective: Brunelleschi's Experiment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Thermal Energy (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner retrieves their cumulative particle model (revised across Lessons 1–5) and makes their final revision to produce their definitive L6 model. The L6 model must show: (1) liquid water molecules at room temperature — close-packed, disordered, in motion; (2) liquid water at 4°C — maximum density, molecules as close as possible; (3) ice at 0°C — hexagonal lattice, greater intermolecular spacing, reduced density; (4) the direction of outward expansion force on the glass walls; (5) the macroscopic result — crack lines in glass. Learners then place their L6 model beside their L1 model (retrieved from their notebooks) and complete a structured written comparison using the frame: 'My Lesson 1 model showed ____ but was incomplete/incorrect because _____. My final model now includes _____ which explains _____ because _____.' Learners write three comparison statements. The lesson closes with each learner writing one 'Transfer Moment' sentence in their notebook: a new situation where the same principles apply — e.g., 'This also explains why farmers in the highlands of Nyandarua cover their water pipes with insulation in January, because if water in the pipes cools below 4°C it expands and can burst the pipe just like our bottle.'	Distribute L1 model sheets (if collected) or ask learners to open notebooks to their Lesson 1 particle diagram. Say: 'Look at your L1 model. This was your best scientific thinking on Day 1. It was not wrong — it was the beginning. Now make your final revision. You have 8 minutes. Add, correct, and annotate. Your L6 model should be the most detailed, accurate, and complete scientific diagram you have ever drawn in this class.' [Set an 8-minute timer. Circulate silently for the first 3 minutes — do not interrupt.] At the 3-minute mark, pose to the whole class without stopping their work: 'Does your model show the hexagonal lattice AND the direction of outward pressure AND the result at the macroscopic level? If any of those three are missing, add them now.' [WAIT TIME: allow them to continue working.] After the 8 minutes: 'Place your L6 model next to your L1 model. Write your three comparison statements using the frame on the board.' [Display the frame. Allow 5 minutes.] Cold-call 3 learners to share one comparison statement each. After each sharing, ask: 'What specific evidence from our lessons caused that change in your thinking?' [WAIT TIME: 10 seconds per learner.] Finally: 'Write your Transfer Moment sentence. This is your gift to yourself — proof that you can use this science outside this classroom. Think of a real place in Kenya, a real material, a real situation. You have 2 minutes.' Cold-call 3 transfer moment sentences. End the lesson by returning to the cracked bottle	Strategy: Cumulative Model Revision with Explicit Before/After Comparison (NGSS SEP 2 — Developing and Using Models). Placing the L1 and L6 models side by side makes conceptual growth visible, concrete, and personally meaningful. The structured comparison frame requires learners to articulate not just what changed but why — connecting model revision to specific evidence. The Transfer Moment sentence is a powerful closure strategy: it tests for genuine understanding (as opposed to performance) by requiring application to a novel, self-generated Kenyan context. This strategy also embeds SEP 6 (Constructing Explanations) and SEP 7 (Engaging in Argument from Evidence) as learners defend their model revisions.	Observable indicator: The L6 particle model contains all five required elements (liquid at room temp, liquid at 4°C, ice lattice with greater spacing, direction of outward force, macroscopic crack result). The three comparison statements explicitly reference specific evidence from the unit. The Transfer Moment sentence identifies a novel Kenyan context, a specific material, and the correct direction of thermal change. Teacher collects all L6 models and three-sentence driving question answers as summative evidence of unit mastery. Flag any learner whose L6 model still does not show the hexagonal lattice with greater spacing than liquid — this indicates the core misconception (water contracts on cooling) has not been resolved and warrants a follow-up conversation before the unit assessment.

		display: 'On Day 1, this bottle puzzled you. Today, you can explain every crack line on it — with a formula, with a particle diagram, and with the knowledge that this same physics shapes how engineers design water pipes in Nairobi, railway tracks on the SGR, and storage tanks in Mombasa. That is the power of physics.'		
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D. TEACHER REFLECTION

After delivering this capstone lesson, reflect on the following questions in your professional journal:

1. **DRIVING QUESTION RESOLUTION:** When the class answered the driving question together in three sentences, how precise and complete were the responses? Could learners articulate all three layers — the macroscopic phenomenon (the crack), the particle-level mechanism (hexagonal lattice with greater spacing), and the quantitative formula ($\Delta V = \beta V_0 \Delta T$)? If any layer was weak, which prior lesson's evidence was insufficiently consolidated?
2. **MODEL GROWTH:** When learners placed their L1 model beside their L6 model, was the growth genuinely visible? Did learners who held the 'things shrink when cold' misconception in Lesson 1 show evidence of having resolved it? How many learners' L6 models still lacked the hexagonal ice lattice diagram — and what does this tell you about the adequacy of Lesson 3's molecular modelling activity?
3. **DQB QUALITY:** Were there DQB questions that no group could resolve — questions that went to orange 'frontier science' notes? If so, were these genuinely frontier questions (e.g., the precise molecular dynamics of the 4°C density maximum) or were they questions that should have been resolved by the unit's content? The latter signals gaps in the instructional sequence to address before next year.
4. **FINAL EXPLANATION POSTER:** During the Poster Parade, what was the quality of peer feedback? Did learners use scientific vocabulary (volumetric expansion, anomalous behaviour, hydrogen bonding, tensile strength) in their feedback, or did they default to vague praise? Consider how to scaffold more precise peer feedback language in future capstone lessons.
5. **KENYAN CONTEXT TRANSFER:** In the Transfer Moment sentences, how many learners generated genuinely local, specific contexts (e.g., water pipes in Nyandarua, water towers in Nairobi's Westlands, fish storage tanks in Kisumu's Lake Victoria waterfront) versus generic statements? Rich local transfer is the hallmark of deep understanding — celebrate and display exemplary Transfer Moment sentences in the classroom.
6. **EQUITY CHECK:** Which learners were most silent during cold-calls? Review your notebook scan data and poster quality for learners who were not cold-called — is there a pattern by gender, by home language, or by prior performance? The CBE framework demands that all learners achieve competency, not just the most vocal. Plan targeted support conversations with any learner whose L6 model still shows core misconceptions before the unit assessment."

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners conducted a six-station Evidence Gallery Walk, revisiting and recording key findings from all five prior lessons — including particle diagrams of liquid water vs. ice, the density-vs-temperature graph showing the 4°C anomaly, and the volumetric expansion formula $\Delta V = \beta V_0 \Delta T$ applied to a 500 ml water bottle. Learners physically placed their Lesson 1 prediction cards beside their final L6 particle models, making conceptual growth visible. The Driving Question Board was returned to the class with all accumulated sticky-note questions, and every question received either a green resolved note or an orange frontier-science note.
What did I learn?	Learners demonstrated that water's anomalous expansion upon freezing is caused by hydrogen bonding, which arranges water molecules into a hexagonal lattice with greater intermolecular spacing than liquid water — reducing density from approximately 1.00 g/cm ³ (liquid at 4°C) to 0.917 g/cm ³ (ice at 0°C). This ~9% volume increase in a sealed, rigid glass bottle generates internal pressure exceeding the tensile strength of glass (33–100 MPa), causing the characteristic crack pattern. Learners applied the volumetric expansion formula to calculate $\Delta V \approx 0.18$ ml for a 500 ml bottle cooling from 4°C to 0°C. They connected this knowledge to Kenyan engineering contexts: burst water pipes in the Kericho and Nyandarua highlands, expansion joints on the SGR Nairobi–Mombasa railway, and storage tank design at the Mombasa port.
How does this explain the phenomenon?	The sealed water bottle cracked because water is one of the very few substances on Earth that expands — rather than contracts — when it freezes. As the temperature dropped from 4°C (the point of maximum density) to 0°C, water molecules formed hydrogen bonds that locked them into a rigid hexagonal lattice with greater spacing between molecules. This expansion increased the volume of water by approximately 9%, generating a pressure force directed outward against the rigid glass walls. Because the glass could not stretch to accommodate this expansion, the stress exceeded the glass's tensile strength and the bottle fractured along its weakest lines. The same principle — quantified by the volumetric expansion formula $\Delta V = \beta V_0 \Delta T$ — explains why engineers build expansion gaps into railway tracks, design flexible couplings in water pipelines, and specify temperature-rated containers for cold storage across Kenya and the world.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.